

# Group & Ring Theory MATP33: Course Summary of Important Results

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## Lecture 0 — Background Results

**Theorem 1** (First Isomorphism Theorem). *Let  $f : G \rightarrow H$  be a group homomorphism. Then  $\ker f \trianglelefteq G$  and  $\operatorname{im} f \cong G/\ker f$ .*

**Theorem 2** (Second Isomorphism Theorem). *Let  $G$  be a group,  $H \leq G$ , and  $N \trianglelefteq G$ . Then  $HN \leq G$ ,  $H \cap N \trianglelefteq H$ , and  $HN/N \cong H/(H \cap N)$ .*

**Theorem 3** (Third Isomorphism Theorem). *If  $K \trianglelefteq G$  and  $K \leq L \trianglelefteq G$ , then  $(G/K)/(L/K) \cong G/L$ .*

**Theorem 4** (Correspondence Theorem). *Let  $N \trianglelefteq G$ . Every subgroup of  $G/N$  is of the form  $H/N$  where  $N \leq H \leq G$ , and conversely if  $N \leq H \leq G$  then  $H/N \leq G/N$ .*

**Theorem 5** (Lagrange). *If  $H \leq G$  and  $G$  is finite, then  $|H|$  divides  $|G|$ .*

**Theorem 6** (Fundamental Theorem of Finitely Generated Abelian Groups). *Let  $A$  be a finitely generated abelian group. Then*

$$A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z},$$

*where  $s \geq 0$  and  $a_1 \mid a_2 \mid \cdots \mid a_r$  are non-zero non-units in  $\mathbb{Z}$ . This decomposition is unique.*

## Lecture 1

**Theorem 7** (1.10). *Let  $G$  be a group and  $X$  a set. Then:*

- (i) If  $X$  is a  $G$ -set, then the action of  $G$  on  $X$  induces a homomorphism  $\varphi : G \rightarrow S_X$ ;*
- (ii) Any homomorphism  $\varphi : G \rightarrow S_X$  induces an action of  $G$  on  $X$ .*

*Proof.* (i) Define  $\varphi : G \rightarrow S_X$  by  $(\varphi(a))(x) = a \cdot x$  for  $a \in G$  and  $x \in X$ . We must first check that  $\varphi(a) \in S_X$  for each  $a \in G$ , i.e. that the function  $x \mapsto a \cdot x$  is a bijection  $X \rightarrow X$ .

- *Injective:* Suppose  $a \cdot x = a \cdot y$ . Then  $x = a^{-1} \cdot (a \cdot x) = a^{-1} \cdot (a \cdot y) = y$ .
- *Surjective:* For any  $x \in X$ , we have  $a \cdot (a^{-1} \cdot x) = x$ , so  $x \in \operatorname{im}(\varphi(a))$ .

Hence  $\varphi(a) \in S_X$ . To see  $\varphi$  is a homomorphism, let  $a, b \in G$ . Then for all  $x \in X$ ,

$$(\varphi(ab))(x) = (ab) \cdot x = a \cdot (b \cdot x) = (\varphi(a))((\varphi(b))(x)) = (\varphi(a)\varphi(b))(x).$$

(ii) Define  $a \cdot x := (\varphi(a))(x)$ . Then  $(ab) \cdot x = (\varphi(ab))(x) = (\varphi(a)\varphi(b))(x) = a \cdot (b \cdot x)$ , and  $e \cdot x = (\varphi(e))(x) = x$ . Hence  $X$  is a  $G$ -set.  $\square$

**Theorem 8** (1.11 — Cayley's Theorem). *Every group  $G$  is isomorphic to a subgroup of the symmetric group  $S_G$ .*

*Proof.* By Theorem 1.10(i), left translation  $a \cdot x := ax$  gives a homomorphism  $\varphi : G \rightarrow S_G$ . If  $a \in \ker \varphi$ , then  $\varphi(a) = \text{id}_G$ , so  $ax = x$  for all  $x \in G$ , giving  $a = e$ . Hence  $\ker \varphi = \{e\}$ , and by the First Isomorphism Theorem  $G \cong G/\ker \varphi \cong \text{im } \varphi \leq S_G$ .  $\square$

**Definition 1** (Normal core). *Let  $G$  be a group and  $H \leq G$ . The normal core of  $H$  in  $G$  is*

$$\text{core}(H) := \bigcap_{x \in G} xHx^{-1}.$$

*It is the largest normal subgroup of  $G$  contained in  $H$ .*

**Theorem 9** (1.12). *Let  $G$  be a group and  $H \leq G$  a subgroup of index  $n$ . Then there is a homomorphism  $\varphi : G \rightarrow S_n$  such that  $\ker \varphi = \text{core}(H)$ .*

*Proof.* Since  $|G/H| = n$ , we have  $S_{G/H} \cong S_n$ . By Theorem 1.10(i), the action of  $G$  on the coset space  $G/H$  by left multiplication gives a homomorphism  $\varphi : G \rightarrow S_n$  with  $\varphi(a)(xH) = axH$ . Then

$$\begin{aligned} \ker \varphi &= \{a \in G \mid axH = xH \text{ for all } x \in G\} \\ &= \{a \in G \mid x^{-1}ax \in H \text{ for all } x \in G\} \\ &= \{a \in G \mid a \in xHx^{-1} \text{ for all } x \in G\} \\ &= \bigcap_{x \in G} xHx^{-1} = \text{core}(H). \end{aligned} \quad \square$$

**Corollary 1** (1.13). *Let  $G$  be a group with a normal subgroup  $H$  of index  $n$ . Then  $G/H$  is isomorphic to a subgroup of  $S_n$ .*

*Proof.* Since  $H \trianglelefteq G$ , conjugation fixes  $H$ , so  $\text{core}(H) = \bigcap_{x \in G} xHx^{-1} = H$ . By Theorem 1.12,  $\ker \varphi = H$ , and by the First Isomorphism Theorem  $G/H \cong \text{im } \varphi \leq S_n$ .  $\square$

**Theorem 10** (1.24 — Orbit-Stabiliser Theorem). *Let  $G$  be a group acting on a set  $X$ . Then the set of all orbits in  $X$  under  $G$  is a partition of  $X$ . Furthermore, for each  $x \in X$  there is a bijection  $Gx \rightarrow G/G_x$ , and hence*

$$|Gx| = |G : G_x|.$$

*Therefore, if  $X$  is finite,*

$$|X| = \sum_{x \in C} |G : G_x|,$$

*where  $C \subseteq X$  contains exactly one element from each orbit.*

*Proof. (Partition.)* Define  $x \sim y$  if  $x = ay$  for some  $a \in G$ . This is an equivalence relation: reflexivity follows from  $e \cdot x = x$ , symmetry from  $x = ay \Rightarrow y = a^{-1}x$ , and transitivity from  $x = ay, y = bz \Rightarrow x = (ab)z$ . The equivalence class of  $x$  is exactly the orbit  $Gx$ , so the orbits partition  $X$ .

(*Bijection*  $Gx \rightarrow G/G_x$ .) Define

$$\varphi : Gx \rightarrow G/G_x, \quad ax \mapsto aG_x.$$

The key observation is the following chain of equivalences, for any  $a, b \in G$ :

$$ax = bx \iff a^{-1}bx = x \iff a^{-1}b \in G_x \iff aG_x = bG_x.$$

- *Well-defined:* If  $ax = bx$  (the same orbit element written using two different group elements), then  $aG_x = bG_x$ , so the coset  $\varphi$  assigns does not depend on the choice of representative.
- *Injective:* If  $\varphi(ax) = \varphi(bx)$ , i.e.  $aG_x = bG_x$ , then  $ax = bx$  by the chain above.
- *Surjective:* Take any coset  $aG_x \in G/G_x$ . Since  $a \in G$ , we have  $ax \in Gx$ , and  $\varphi(ax) = aG_x$ .

Hence  $\varphi$  is a bijection, giving  $|Gx| = |G/G_x| = |G : G_x|$ . Summing over one representative per orbit gives  $|X| = \sum_{x \in C} |G : G_x|$ .  $\square$

**Theorem 11** (1.25 — Class Equation). *Let  $G$  be a finite group. Then the conjugacy classes partition  $G$ , and*

$$|G| = \sum_{a \in C} |G : C_G(a)|,$$

where  $C \subseteq G$  contains exactly one element from each conjugacy class.

*Proof.* Let  $G$  act on itself by conjugation, i.e.  $a \cdot x = axa^{-1}$ . The orbit of  $x$  under this action is its conjugacy class  $\text{ccl}(x) = \{axa^{-1} \mid a \in G\}$ , and the stabiliser of  $x$  is  $G_x = \{a \in G \mid axa^{-1} = x\} = C_G(x)$ . Applying Theorem 1.24 directly gives the partition into orbits and  $|G| = \sum_{a \in C} |G : C_G(a)|$ .  $\square$

## Lecture 2

**Theorem 12** (2.1). *Let  $G$  be a finite group of order  $p^n$ , where  $p$  is prime and  $n \in \mathbb{N}$ . Then:*

- (i)  $G$  has a non-trivial centre  $Z(G)$ ;
- (ii)  $Z(G) \cap K$  is non-trivial for any non-trivial normal subgroup  $K$  of  $G$ ;
- (iii) if  $H$  is a proper subgroup of  $G$ , then  $H \subsetneq N_G(H)$ . In particular, if  $|H| = p^{n-1}$  then  $H \trianglelefteq G$ .

*Proof.* (i) Since  $x \in Z(G)$  iff  $\text{ccl}(x) = \{x\}$ , the class equation refines to

$$p^n = |Z(G)| + \sum_{x \in C} |G : C_G(x)|,$$

where  $C$  runs over conjugacy classes with more than one element. For each  $x \in C$ ,  $|C_G(x)|$  properly divides  $p^n$ , so  $p \nmid |G : C_G(x)|$ . Hence  $p$  divides the sum, and therefore  $p \mid |Z(G)|$ . Since  $e \in Z(G)$ , we have  $|Z(G)| \geq 1$ , so in fact  $|Z(G)| \geq p > 1$ .

(ii) Intersecting the class decomposition of  $G$  with  $K$  gives  $|K| = |Z(G) \cap K| + \sum_{x \in C} |\text{ccl}(x) \cap K|$ . Since  $K \trianglelefteq G$ , each conjugacy class is either disjoint from  $K$  or contained in  $K$ , so  $p \mid \sum |\text{ccl}(x) \cap K|$ . As  $p \mid |K|$ , we get  $p \mid |Z(G) \cap K|$ , and the same argument as (i) gives  $Z(G) \cap K$  non-trivial.

(iii) Let  $M$  be a maximal normal subgroup of  $G$  contained in  $H$ . Then  $G/M$  is a  $p$ -group, so by (i) it has a non-trivial centre  $L/M \trianglelefteq G/M$ . By the Correspondence Theorem,  $L \trianglelefteq G$ . Since  $M \subsetneq L$  strictly, maximality of  $M$  forces  $L \not\subseteq H$ .

It remains to show  $L \subseteq N_G(H)$ . To show  $\ell \in N_G(H)$  we need  $\ell^{-1}h\ell \in H$  for all  $h \in H$ . Take  $h \in H$  and  $\ell \in L$ . Since  $\ell M \in L/M = Z(G/M)$ , it commutes with every coset, so  $(\ell M)(hM) = (hM)(\ell M)$ , i.e.  $\ell hM = h\ell M$ . This gives  $\ell^{-1}h\ell \in hM$ . Since  $h \in H$  and  $M \subseteq H$ , we have  $hM \subseteq H$ , so  $\ell^{-1}h\ell \in H$ . Hence  $\ell \in N_G(H)$ , so  $L \subseteq N_G(H)$ .

Since  $L \subseteq N_G(H)$  but  $L \not\subseteq H$ , we have  $H \subsetneq N_G(H)$ . Finally, if  $|H| = p^{n-1}$  then  $N_G(H)$  has order strictly greater than  $p^{n-1}$ , and the only possibility is  $p^n = |G|$ , so  $N_G(H) = G$  and  $H \trianglelefteq G$ .  $\square$

**Theorem 13** (2.2 — Burnside's Lemma). *Let  $G$  be a finite group acting on a finite set  $X$ , and for  $g \in G$  let  $X_g := \{x \in X \mid gx = x\}$ . Then the number  $k$  of orbits in  $X$  under  $G$  is*

$$k = \frac{1}{|G|} \sum_{g \in G} |X_g|.$$

*Proof.* Let  $S := \{(g, x) \in G \times X \mid gx = x\}$ . Fixing  $g$  gives  $|S| = \sum_{x \in X} |X_g|$ ; fixing  $x$  gives  $|S| = \sum_{g \in G} |G_x|$ . Let  $C \subseteq X$  contain one element from each orbit. By the Orbit-Stabiliser Theorem,  $|G_x| = |G|/|Gx| = |G|/|Ga|$  for all  $x \in Ga$ . Hence

$$\frac{1}{|G|} \sum_{g \in G} |X_g| = \frac{|S|}{|G|} = \frac{1}{|G|} \sum_{x \in X} |G_x| = \frac{1}{|G|} \sum_{a \in C} \sum_{x \in Ga} \frac{|G|}{|Ga|} = \frac{1}{|G|} \sum_{a \in C} |G| = k. \quad \square$$

**Remark 1** (Pólya counting). *Burnside's Lemma counts distinct objects under symmetry. As an illustration: how many necklaces with  $p$  beads ( $p$  prime) can be made from  $n$  colours, where two necklaces are the same if one is a rotation of the other?*

*Let  $X$  be the set of all colourings, so  $|X| = n^p$ . The cyclic group  $G = \langle a \rangle$  of order  $p$  acts on  $X$  by rotation. For the identity,  $X_e = X$  so  $|X_e| = n^p$ . For any  $g \neq e$ : since  $|G|$  is prime,  $g$  generates  $G$ , so a necklace fixed by  $g$  must be fixed by all rotations, meaning all beads have the same colour. Hence  $|X_g| = n$ . Burnside then gives*

$$k = \frac{1}{p} (n^p + (p-1)n).$$

## Lecture 3

**Definition 2** ( $p$ -group, Sylow  $p$ -subgroup). *Let  $p$  be a prime. A group  $G$  is a  $p$ -group if every element of  $G$  has order a power of  $p$ . If  $G$  is finite and  $p^m \mid |G|$  but  $p^{m+1} \nmid |G|$ , then any subgroup of  $G$  of order  $p^m$  is called a Sylow  $p$ -subgroup of  $G$ .*

**Theorem 14** (3.5 — First Sylow Theorem). *Let  $G$  be a finite group and  $p$  a prime. If  $p^m \mid |G|$ , then  $G$  has a subgroup of order  $p^m$ .*

*Proof.* If  $m = 0$  the result is trivial. We proceed by induction on  $n = |G|$ , the case  $n = 1$  being trivial. Assume the result holds for all groups of order less than  $n$ .

*Case 1:*  $p \mid |Z(G)|$ . By the Fundamental Theorem of Finitely Generated Abelian Groups,  $Z(G)$  contains an element  $a$  of order  $p$ . Let  $C = \langle a \rangle$ . Since  $a \in Z(G)$ , we have  $C \trianglelefteq G$ , and  $|G/C| = n/p$ , which is divisible by  $p^{m-1}$ . By the induction hypothesis,  $G/C$  has a subgroup of order  $p^{m-1}$ , which by the Correspondence Theorem is of the form  $H/C$  for some  $H \leq G$ . Then  $|H| = |H/C| \cdot |C| = p^{m-1} \cdot p = p^m$ .

*Case 2:*  $p \nmid |Z(G)|$ . From the class equation  $n = |Z(G)| + \sum_a |G : C_G(a)|$ , since  $p \mid n$  but  $p \nmid |Z(G)|$ , there must exist some  $a \notin Z(G)$  with  $p \nmid |G : C_G(a)|$ . Since  $|G| = |G : C_G(a)| \cdot |C_G(a)|$  and  $p^m \mid |G|$  but  $p \nmid |G : C_G(a)|$ , it follows that  $p^m \mid |C_G(a)|$ . Since  $a \notin Z(G)$ , we have  $C_G(a) \neq G$ , so  $|C_G(a)| < n$ . By the induction hypothesis  $C_G(a)$  contains a subgroup of order  $p^m$ , which is also a subgroup of  $G$ .  $\square$

**Corollary 2** (3.6 — Cauchy's Theorem). *If  $p$  is a prime dividing  $|G|$ , then  $G$  has an element of order  $p$ .*

*Proof.* By the First Sylow Theorem,  $G$  has a subgroup  $H$  of order  $p$ . Every non-identity element of  $H$  has order  $p$ .  $\square$

**Corollary 3** (3.7). *A finite group  $G$  is a  $p$ -group if and only if  $|G|$  is a power of  $p$ .*

*Proof.* If  $|G|$  is a power of  $p$ , then by Lagrange every element order divides  $|G|$ , hence is a power of  $p$ . Conversely, if  $|G|$  is not a power of  $p$ , then some prime  $q \neq p$  divides  $|G|$ , and by Cauchy's Theorem  $G$  has an element of order  $q$ , contradicting that  $G$  is a  $p$ -group.  $\square$

**Theorem 15** (3.8 — Second Sylow Theorem). *Let  $G$  be a finite group and  $p$  a prime. All Sylow  $p$ -subgroups of  $G$  are conjugate.*

*Proof.* Let  $S, H \in \text{Syl}_p(G)$ . Let  $S$  act on the coset space  $G/H$  by left translation:  $s \cdot gH = sgH$ .

*Orbits have  $p$ -power size.* By the Orbit-Stabiliser Theorem, each orbit has size  $|S : S_{gH}|$ , which divides  $|S| = p^m$ , hence is a power of  $p$ .

*There must be a fixed point.* Since  $H$  is a Sylow  $p$ -subgroup,  $p \nmid |G|/|H| = |G/H|$ . Writing  $|G/H|$  as a sum of orbit sizes, each of which is a power of  $p$ , the only way this sum is not divisible by  $p$  is if at least one orbit has size  $p^0 = 1$ .

*A fixed point gives conjugacy.* If  $gH$  is a fixed point, then  $s \cdot gH = gH$  for all  $s \in S$ , so  $g^{-1}sg \in H$  for all  $s \in S$ . Hence  $g^{-1}Sg \subseteq H$ , and since  $|g^{-1}Sg| = |S| = |H|$ , we get  $g^{-1}Sg = H$ . So  $S$  and  $H$  are conjugate.  $\square$

**Corollary 4** (3.10). *A Sylow  $p$ -subgroup of  $G$  is unique if and only if it is normal in  $G$ .*

*Proof.* If  $P$  is the unique Sylow  $p$ -subgroup, then  $gPg^{-1} = P$  for all  $g \in G$  (since conjugates are also Sylow  $p$ -subgroups), so  $P \trianglelefteq G$ . Conversely, if  $P \trianglelefteq G$  then  $P$  is its own conjugate, so by the Second Sylow Theorem it is the only Sylow  $p$ -subgroup.  $\square$

**Theorem 16** (3.9 — Third Sylow Theorem). *Let  $G$  be a finite group and  $p$  a prime. The number  $n_p$  of Sylow  $p$ -subgroups satisfies  $n_p \mid |G|$  and  $n_p \equiv 1 \pmod{p}$ .*

*Proof.*  $n_p \mid |G|$ . By the Second Sylow Theorem, all Sylow  $p$ -subgroups are conjugate to a fixed  $K \in \text{Syl}_p(G)$ , so  $n_p = |\text{ccl}(K)| = |G : N_G(K)|$ , which divides  $|G|$ .

$n_p \equiv 1 \pmod{p}$ . Let  $P \in \text{Syl}_p(G)$ . Since conjugates of Sylow  $p$ -subgroups are again Sylow  $p$ -subgroups (by the Second Sylow Theorem),  $P$  acts on the set  $\text{Syl}_p(G)$  by conjugation:

$$x \cdot Q := xQx^{-1}, \quad x \in P, Q \in \text{Syl}_p(G).$$

This action partitions  $\text{Syl}_p(G)$  into orbits. Let  $F \subseteq \text{Syl}_p(G)$  denote the set of fixed points (those  $Q$  with  $xQx^{-1} = Q$  for all  $x \in P$ ), and let  $\mathcal{O}$  denote the collection of orbits of size greater than 1. Then:

$$n_p = |\text{Syl}_p(G)| = |F| + \sum_{O \in \mathcal{O}} |O|.$$

By the Orbit-Stabiliser Theorem each  $|O|$  divides  $|P| = p^m$ , so every  $O \in \mathcal{O}$  has size  $p^e$  for some  $e \geq 1$ , hence  $p \mid |O|$ . Therefore  $n_p \equiv |F| \pmod{p}$ .

It remains to show  $F = \{P\}$ , which gives  $n_p \equiv 1 \pmod{p}$ . Clearly  $P \in F$ . If  $Q \in F$ , then  $xQx^{-1} = Q$  for all  $x \in P$ , so  $P \leq N_G(Q)$ . By Corollary 3.10, since  $Q \trianglelefteq N_G(Q)$ ,  $Q$  is the unique Sylow  $p$ -subgroup of  $N_G(Q)$ . But  $P$  is also a Sylow  $p$ -subgroup of  $N_G(Q)$ , so  $P = Q$ . Hence  $F = \{P\}$ .  $\square$

## Lecture 4 — Modules: the Basics

**Definition 3** (4.1 — Left  $R$ -module). *Let  $R$  be a ring. An abelian group  $M$  is a left  $R$ -module if there is a map  $R \times M \rightarrow M$ ,  $(r, m) \mapsto rm$ , such that for all  $r, r_1, r_2 \in R$  and  $m, m_1, m_2 \in M$ :*

- (i)  $r(m_1 + m_2) = rm_1 + rm_2$ ;
- (ii)  $(r_1 + r_2)m = r_1m + r_2m$ ;
- (iii)  $(r_1r_2)m = r_1(r_2m)$ ;
- (iv)  $1m = m$  if  $1 \in R$ .

**Lemma 1** (4.7). *Let  $M$  be an  $R$ -module. Then for  $a \in R$  and  $m \in M$ :*

- (i)  $0m = 0$ ;
- (ii)  $a0 = 0$ ;
- (iii)  $(-a)m = -(am) = a(-m)$ .

*Proof.* (i)  $0m = (0 + 0)m = 0m + 0m$ . Subtracting  $0m$  from both sides gives  $0 = 0m$ .

(ii)  $a0 = a(0 + 0) = a0 + a0$ . Subtracting  $a0$  from both sides gives  $0 = a0$ .

(iii) By (i) and axiom (ii):  $0 = 0m = (a + (-a))m = am + (-a)m$ , so  $(-a)m = -(am)$ . By (ii) and axiom (i):  $0 = a0 = a(m + (-m)) = am + a(-m)$ , so  $a(-m) = -(am)$ .  $\square$

**Definition 4** (4.8 — Submodule). *A subset  $N$  of an  $R$ -module  $M$  is an  $R$ -submodule of  $M$  if:*

- (i)  $N \neq \emptyset$ ;
- (ii)  $a - b \in N$  for all  $a, b \in N$ ;
- (iii)  $ra \in N$  for all  $a \in N, r \in R$ .

**Theorem 17** (4.12). *Let  $(N_i)_{i \in I}$  be a family of  $R$ -submodules of an  $R$ -module  $M$ . Then  $\bigcap_{i \in I} N_i$  is also an  $R$ -submodule of  $M$ .*

*Proof.* Since  $0 \in N_i$  for all  $i$ , we have  $0 \in \bigcap N_i$ , so it is non-empty. Let  $x, y \in \bigcap N_i$  and  $r \in R$ . For each  $i$ , since  $N_i$  is a submodule,  $x - y \in N_i$  and  $rx \in N_i$ . Hence  $x - y$  and  $rx$  lie in  $\bigcap N_i$ .  $\square$

**Definition 5** (4.13 — Generated submodule, cyclic module). *Let  $S \subseteq M$ . The family  $\mathcal{A} = \{N \mid N \text{ is an } R\text{-submodule of } M \text{ containing } S\}$  is non-empty (since  $M \in \mathcal{A}$ ), so by Theorem 4.12 the intersection  $\bigcap_{N \in \mathcal{A}} N$  is an  $R$ -submodule. It is the smallest submodule of  $M$  containing  $S$ , called the submodule generated by  $S$  and denoted  $(S)$ . If  $S = \{x_1, \dots, x_n\}$  we write  $(x_1, \dots, x_n)$ . An  $R$ -module  $M$  is cyclic if  $M = (x)$  for some  $x \in M$ .*

**Theorem 18** (4.15). *Let  $R$  be a ring with unity and  $x_1, \dots, x_n \in M$ . Then*

$$(x_1, \dots, x_n) = \sum_{i=1}^n Rx_i := \{r_1x_1 + \dots + r_nx_n \mid r_i \in R\}.$$

*Proof.*  $(\supseteq)$  The submodule  $(x_1, \dots, x_n)$  contains each  $x_i$ . By closure under scalar multiplication it contains  $rx_i$  for all  $r \in R$ , and by closure under addition it contains every sum  $r_1x_1 + \dots + r_nx_n$ . Hence  $\sum Rx_i \subseteq (x_1, \dots, x_n)$ .

$(\subseteq)$  We show  $\sum Rx_i$  is a submodule containing each  $x_i$ ; then  $(x_1, \dots, x_n) \subseteq \sum Rx_i$  follows from minimality of  $(x_1, \dots, x_n)$ .

Each  $x_i \in \sum Rx_i$  since  $x_i = 1 \cdot x_i$  (using unity). For the submodule conditions:  $0 = \sum 0 \cdot x_i \in \sum Rx_i$ , so it is non-empty. For  $\sum r_ix_i, \sum s_ix_i \in \sum Rx_i$  and  $r \in R$ ,

$$\sum r_ix_i - \sum s_ix_i = \sum (r_i - s_i)x_i \in \sum Rx_i, \quad r \sum r_ix_i = \sum (rr_i)x_i \in \sum Rx_i. \quad \square$$

**Definition 6** (4.18 — Sum of submodules). *Let  $N_1, \dots, N_k$  be  $R$ -submodules of  $M$ . Their sum  $\sum_{i=1}^k N_i$  is the submodule of  $M$  generated by  $\bigcup_{i=1}^k N_i$ , i.e. the smallest submodule of  $M$  containing each  $N_i$ .*

## Lecture 5 — Modules: continuation

**Theorem 19** (4.19). *Let  $N_1, \dots, N_k$  be  $R$ -submodules of an  $R$ -module  $M$ . Then*

$$\sum_{i=1}^k N_i = \{x_1 + \dots + x_k \mid x_i \in N_i\}.$$

*Proof.* Let

$$A = \{x_1 + \dots + x_k \mid x_i \in N_i\}.$$

It is straightforward to check that  $A$  is a submodule of  $M$ , and it contains every  $N_i$ . Hence, by minimality,

$$\sum_{i=1}^k N_i \subseteq A.$$

Conversely,  $\sum_{i=1}^k N_i$  contains every  $N_i$  and is closed under addition, so it contains every element of  $A$ . Therefore

$$A \subseteq \sum_{i=1}^k N_i.$$

□

**Definition 7** (4.21 — Direct sum). *The sum  $N_1 + \cdots + N_k$  is direct if every element has a unique representation*

$$x = x_1 + \cdots + x_k, \quad x_i \in N_i.$$

In this case we write

$$N_1 \oplus \cdots \oplus N_k.$$

**Theorem 20** (4.22). *Let  $N_1, \dots, N_k$  be submodules of an  $R$ -module  $M$ . The following are equivalent:*

- (i)  $\sum_{i=1}^k N_i$  is direct;
- (ii) if  $\sum_{i=1}^k x_i = 0$ , where  $x_i \in N_i$ , then every  $x_i = 0$ ;
- (iii) for every  $i$ ,

$$N_i \cap \sum_{j \neq i} N_j = \{0\}.$$

*Proof.* (i)  $\Rightarrow$  (ii): Zero already has the representation  $0 + \cdots + 0$ , so uniqueness implies that every  $x_i = 0$ .

(ii)  $\Rightarrow$  (iii): If

$$x \in N_i \cap \sum_{j \neq i} N_j,$$

then  $x = \sum_{j \neq i} x_j$  for some  $x_j \in N_j$ . Hence

$$x - \sum_{j \neq i} x_j = 0,$$

so (ii) gives  $x = 0$ .

(iii)  $\Rightarrow$  (i): Suppose

$$\sum_{i=1}^k x_i = \sum_{i=1}^k y_i, \quad x_i, y_i \in N_i.$$

For every  $i$ ,

$$x_i - y_i = - \sum_{j \neq i} (x_j - y_j) \in N_i \cap \sum_{j \neq i} N_j = \{0\}.$$

Thus  $x_i = y_i$  for every  $i$ , so the representation is unique.

□



**Definition 8** (4.23 — *R*-homomorphism). Let  $M$  and  $N$  be  $R$ -modules. A map  $f: M \rightarrow N$  is an  $R$ -homomorphism if, for all  $x, y \in M$  and  $r \in R$ ,

$$f(x + y) = f(x) + f(y), \quad f(rx) = rf(x).$$

The set of all  $R$ -homomorphisms  $M \rightarrow N$  is denoted

$$\text{Hom}_R(M, N).$$

An  $R$ -homomorphism  $M \rightarrow M$  is called an endomorphism, and

$$\text{End}_R(M) := \text{Hom}_R(M, M).$$

**Theorem 21** (4.30). Let  $M = \bigoplus_{i=1}^k M_i$  be a direct sum of  $R$ -modules. Then

$$\text{Hom}_R(M, M) \cong (\text{Hom}_R(M_j, M_i))_{1 \leq i, j \leq k}$$

as rings, where the right-hand side is a ring  $T$  of  $k \times k$  matrices under the usual matrix addition and multiplication, with  $(i, j)$  entry in  $\text{Hom}_R(M_j, M_i)$ .

*Proof.* Let  $\lambda_j: M_j \rightarrow M$  be the natural inclusion and  $\pi_i: M \rightarrow M_i$  the natural projection. Two key identities hold:

$$\sum_{\ell=1}^k \lambda_\ell \pi_\ell = \text{id}_M, \quad \pi_p \lambda_q = \delta_{pq} \text{id}_{M_p}.$$

The first says every  $x \in M$  equals the sum of its components; the second says projecting the  $q$ -th inclusion gives zero unless  $p = q$ .

Define  $\sigma: \text{Hom}_R(M, M) \rightarrow T$  by  $\sigma(\phi) = (\pi_i \phi \lambda_j)_{ij}$ .

*Ring homomorphism.* Additivity is immediate. For multiplication, using the first identity:

$$(\sigma(\phi)\sigma(\psi))_{ij} = \sum_{\ell} \pi_i \phi \lambda_\ell \pi_\ell \psi \lambda_j = \pi_i \phi \left( \sum_{\ell} \lambda_\ell \pi_\ell \right) \psi \lambda_j = \pi_i \phi \text{id}_M \psi \lambda_j = (\sigma(\phi\psi))_{ij}.$$

*Injective.* Suppose  $\sigma(\phi) = 0$ , so  $\pi_i \phi \lambda_j = 0$  for all  $i, j$ . Applying  $\lambda_i$  on the left and summing over  $i$  using the first identity:

$$\phi \lambda_j = \text{id}_M \cdot \phi \lambda_j = \left( \sum_i \lambda_i \pi_i \right) \phi \lambda_j = \sum_i \lambda_i (\pi_i \phi \lambda_j) = 0.$$

Now applying  $\pi_j$  on the right and summing over  $j$ :

$$\phi = \phi \cdot \text{id}_M = \phi \left( \sum_j \lambda_j \pi_j \right) = \sum_j (\phi \lambda_j) \pi_j = 0.$$

*Surjective.* Given  $f = (f_{ij}) \in T$ , set  $\phi = \sum_{i,j} \lambda_i f_{ij} \pi_j$ . Using the second identity:

$$(\sigma(\phi))_{st} = \pi_s \phi \lambda_t = \sum_{i,j} \pi_s \lambda_i f_{ij} \pi_j \lambda_t = \sum_{i,j} \delta_{si} f_{ij} \delta_{jt} = f_{st}.$$

Hence  $\sigma(\phi) = f$ , so  $\sigma$  is surjective. □

**Definition 9** (Quotient module). Let  $N$  be an  $R$ -submodule of an  $R$ -module  $M$ . The set of cosets  $M/N = \{a + N \mid a \in M\}$  becomes an  $R$ -module under

$$(a + N) + (b + N) = (a + b) + N, \quad r(a + N) = ra + N.$$

It is called the quotient module of  $M$  by  $N$ .

**Theorem 22** (4.31 — Correspondence Theorem for Modules). Let  $N$  be an  $R$ -submodule of an  $R$ -module  $M$ . Every  $R$ -submodule of  $M/N$  is of the form  $U/N$ , where  $U$  is an  $R$ -submodule of  $M$  containing  $N$ .

*Proof.* The canonical projection  $f : M \rightarrow M/N$ ,  $f(x) = x + N$ , is an  $R$ -homomorphism:

$$f(x+y) = (x+y)+N = (x+N)+(y+N) = f(x)+f(y), \quad f(rx) = rx+N = r(x+N) = rf(x).$$

Let  $X$  be an  $R$ -submodule of  $M/N$ . The structure of the proof is: define  $U$  as the preimage of  $X$  under  $f$ , show  $U$  is a submodule of  $M$  containing  $N$ , and conclude  $X = U/N$ .

Set  $U = f^{-1}(X) = \{x \in M \mid f(x) \in X\}$ .

$N \subseteq U$ . For any  $x \in N$ ,  $f(x) = x + N = 0 + N = 0_{M/N} \in X$  (since  $X$  is a submodule, it contains  $0$ ). Hence  $x \in U$ .

$U$  is an  $R$ -submodule. Since  $0 \in N \subseteq U$ ,  $U$  is non-empty. For  $x, y \in U$  and  $r \in R$ , since  $f$  is an  $R$ -homomorphism:

$$f(x - y) = f(x) - f(y) \in X, \quad f(rx) = rf(x) \in X,$$

so  $x - y \in U$  and  $rx \in U$ .

$X = U/N$ . By definition of  $U$ ,  $f(U) \subseteq X$ . For the reverse inclusion, take any  $x \in X$ ; since  $f$  is surjective there exists  $y \in M$  with  $f(y) = x$ , so  $y \in U$  and  $x \in f(U)$ . Hence  $X = f(U) = U/N$ .  $\square$

**Theorem 23** (4.32 — First Isomorphism Theorem for Modules). Let  $f : M \rightarrow N$  be an  $R$ -homomorphism. Then  $M/\ker f \cong f(M)$ .

*Proof.* Define  $g : M/\ker f \rightarrow f(M)$  by  $g(x + \ker f) = f(x)$ . We know from group theory that  $g$  is an isomorphism of abelian groups (well-defined and injective because  $x + \ker f = y + \ker f \iff x - y \in \ker f \iff f(x) = f(y)$ ; surjective onto  $f(M)$  by definition). It remains to check that  $g$  respects the  $R$ -action:

$$g(r(x + \ker f)) = g(rx + \ker f) = f(rx) = rf(x) = rg(x + \ker f). \quad \square$$

**Theorem 24** (Second Isomorphism Theorem for Modules). Let  $U$  and  $V$  be  $R$ -submodules of an  $R$ -module  $M$ . Then  $(U + V)/V \cong U/(U \cap V)$ .

*Proof.* Define  $f : U \rightarrow (U + V)/V$  by  $f(u) = u + V$ .

$f$  is an  $R$ -homomorphism. For  $u_1, u_2 \in U$  and  $r \in R$ :

$$f(u_1 + u_2) = (u_1 + u_2) + V = (u_1 + V) + (u_2 + V) = f(u_1) + f(u_2),$$

$$f(ru) = ru + V = r(u + V) = rf(u).$$

$f$  is surjective. Any element of  $(U + V)/V$  has the form  $(u + v) + V = u + V = f(u)$  for some  $u \in U$ ,  $v \in V$  (since  $v \in V$  means  $v + V = V$ ).

$\ker f = U \cap V$ . We have  $f(u) = 0$  iff  $u + V = V$  iff  $u \in V$ , so  $\ker f = \{u \in U \mid u \in V\} = U \cap V$ .

By the First Isomorphism Theorem,  $U/(U \cap V) \cong (U + V)/V$ .  $\square$

**Theorem 25** (Third Isomorphism Theorem for Modules). *Let  $U \subseteq V$  be  $R$ -submodules of an  $R$ -module  $M$ . Then  $(M/U)/(V/U) \cong M/V$ .*

*Proof.* Define  $f : M/U \rightarrow M/V$  by  $f(x+U) = x+V$ .

$f$  is well-defined. If  $x+U = y+U$  then  $x-y \in U \subseteq V$ , so  $x+V = y+V$ .

$f$  is an  $R$ -homomorphism. For  $x+U, y+U \in M/U$  and  $r \in R$ :

$$f((x+U)+(y+U)) = f((x+y)+U) = (x+y)+V = (x+V)+(y+V) = f(x+U)+f(y+U),$$

$$f(r(x+U)) = f(rx+U) = rx+V = r(x+V) = rf(x+U).$$

$f$  is surjective. For any  $x+V \in M/V$ ,  $f(x+U) = x+V$ .

$\ker f = V/U$ . We have  $f(x+U) = 0$  iff  $x+V = V$  iff  $x \in V$ , so  $\ker f = \{x+U \mid x \in V\} = V/U$ .

By the First Isomorphism Theorem,  $(M/U)/(V/U) \cong M/V$ .  $\square$

**Theorem 26** (4.33). *Let  $A \leq M$  and  $B \leq N$  be  $R$ -submodules. Then  $(M \times N)/(A \times B) \cong M/A \times N/B$ .*

*Proof.* Define  $f : M \times N \rightarrow M/A \times N/B$  by  $f(m, n) = (m+A, n+B)$ . This is a surjective  $R$ -homomorphism, and  $\ker f = \{(m, n) \mid m \in A, n \in B\} = A \times B$ . By the First Isomorphism Theorem,  $(M \times N)/(A \times B) \cong M/A \times N/B$ .  $\square$

## Lecture 6 — Simple, Completely Reducible, and Free Modules

**Definition 10** (5.1 — Simple module). *Let  $R$  be a ring with unity. An  $R$ -module  $M$  is called simple (or irreducible) if  $M \neq 0$  and the only  $R$ -submodules of  $M$  are  $0$  and  $M$ .*

**Theorem 27** (5.5 — Characterisation of Simple Modules). *Let  $R$  be a ring with unity and  $M$  an  $R$ -module. The following are equivalent:*

- (i)  $M$  is simple;
- (ii)  $M \neq 0$ , and  $M$  is generated by any  $0 \neq x \in M$ ;
- (iii)  $M \cong R/I$ , where  $I$  is a maximal left ideal of  $R$ .

*Proof.* (i)  $\Rightarrow$  (ii): Let  $0 \neq x \in M$ . Then  $Rx$  is a non-zero submodule of  $M$ , so  $Rx = M$  by simplicity.

(ii)  $\Rightarrow$  (iii): Take  $0 \neq x \in M$  and define  $f : R \rightarrow M$  by  $f(r) = rx$ . This is a surjective  $R$ -homomorphism, so letting  $I = \ker f$ , the First Isomorphism Theorem gives  $R/I \cong M$ . Note that (ii) implies  $M$  is simple: if  $0 \neq N \leq M$  and  $0 \neq x \in N$ , then  $M = Rx \subseteq N$ . Hence  $R/I \cong M$  is simple. By the Correspondence Theorem (4.31), any left ideal  $J$  with  $I \subseteq J \subseteq R$  corresponds to a submodule  $J/I$  of  $R/I$ . Simplicity forces  $J/I = 0$  or  $J/I = R/I$ , i.e.  $J = I$  or  $J = R$ , so  $I$  is maximal.

(iii)  $\Rightarrow$  (i): Let  $0 \neq N \leq M \cong R/I$ . By the Correspondence Theorem (4.31),  $N \cong J/I$  for some left ideal  $J$  with  $I \subsetneq J \subseteq R$ . Since  $I$  is maximal,  $J = R$ , so  $N = M$ .  $\square$

**Theorem 28** (5.6 — Schur's Lemma). *Let  $M$  be a simple  $R$ -module. Then  $\text{Hom}_R(M, M)$  is a division ring.*

*Proof.* Let  $0 \neq f \in \text{Hom}_R(M, M)$ . We show  $f$  is invertible and  $f^{-1} \in \text{Hom}_R(M, M)$ .

$f$  is bijective. Since  $\ker f$  and  $\text{im } f$  are submodules of  $M$ , simplicity forces each to be 0 or  $M$ . As  $f \neq 0$ , we have  $\ker f \neq M$  and  $\text{im } f \neq 0$ , so  $\ker f = 0$  (injective) and  $\text{im } f = M$  (surjective).

$f^{-1}$  is an  $R$ -homomorphism. We cannot push  $f^{-1}$  through operations directly, but we can push  $f$  through them. The strategy is: to prove  $f^{-1}(u) = v$ , it suffices to show  $f(u) = f(v)$ , since  $f$  is injective. For  $x, y \in M$  and  $r \in R$ :

$$\begin{aligned} f(f^{-1}(x + y)) &= x + y = f(f^{-1}(x)) + f(f^{-1}(y)) = f(f^{-1}(x) + f^{-1}(y)), \\ f(f^{-1}(rx)) &= rx = rf(f^{-1}(x)) = f(rf^{-1}(x)), \end{aligned}$$

where we used that  $f$  is itself an  $R$ -homomorphism in the last step of each line. Injectivity of  $f$  gives  $f^{-1}(x + y) = f^{-1}(x) + f^{-1}(y)$  and  $f^{-1}(rx) = rf^{-1}(x)$ .  $\square$

**Definition 11** (5.7 — Completely reducible module). Let  $R$  be a ring with unity. An  $R$ -module  $M$  is called completely reducible (or semisimple) if  $M = \sum_{\alpha \in \Lambda} M_\alpha$ , where each  $M_\alpha$  is a simple  $R$ -submodule.

**Theorem 29** (5.9). Let  $M = \sum_{\alpha \in \Lambda} M_\alpha$  be a sum of simple  $R$ -submodules, and let  $K$  be a submodule of  $M$ . Then there exists  $\Lambda' \subseteq \Lambda$  such that  $M = K \oplus (\bigoplus_{\alpha \in \Lambda'} M_\alpha)$ .

*Proof.* Let  $S$  be the family of subsets  $A \subseteq \Lambda$  satisfying two conditions:  $\bigoplus_{\alpha \in A} M_\alpha$  is a direct sum, and  $K \cap \bigoplus_{\alpha \in A} M_\alpha = 0$ . Together these mean  $K \oplus \bigoplus_{\alpha \in A} M_\alpha$  is a genuine direct sum. Since  $\emptyset$  satisfies both conditions trivially,  $S \neq \emptyset$ . Partially order  $S$  by inclusion; every chain  $(A_i)$  in  $S$  has upper bound  $\bigcup A_i$  in  $S$ . By Zorn's Lemma,  $S$  has a maximal element  $A$ .

Set  $N = K \oplus \bigoplus_{\alpha \in A} M_\alpha$ . We claim  $N = M$ . Fix any  $\beta \in \Lambda$ . Since  $M_\beta$  is simple,  $M_\beta \cap N$  is either 0 or  $M_\beta$ . If  $M_\beta \cap N = 0$ , then  $M_\beta$  intersects neither  $K$  nor  $\bigoplus_{\alpha \in A} M_\alpha$ , so  $A \cup \{\beta\}$  would satisfy both conditions in  $S$ , contradicting maximality of  $A$ . Hence  $M_\beta \cap N = M_\beta$ , i.e.  $M_\beta \subseteq N$ , for every  $\beta \in \Lambda$ . Since  $M = \sum_{\beta} M_\beta$ , we conclude  $N = M$ .  $\square$

**Corollary 5** (5.10). If  $M = \sum_{\alpha \in \Lambda} M_\alpha$  is a sum of simple  $R$ -submodules, then there exists  $\Lambda' \subseteq \Lambda$  such that  $M = \bigoplus_{\alpha \in \Lambda'} M_\alpha$ .

*Proof.* Apply Theorem 5.9 with  $K = \{0\}$ .  $\square$

**Definition 12** (5.11 — Linear independence and basis). A sequence  $x_1, \dots, x_n$  in an  $R$ -module  $M$  is linearly independent if  $\sum_{i=1}^n a_i x_i = 0$  implies  $a_1 = \dots = a_n = 0$  for any  $a_i \in R$ . A subset  $S \subseteq M$  is linearly independent if every finite sequence of distinct elements of  $S$  is linearly independent.

A subset  $B \subseteq M$  is a basis of  $M$  if  $B$  generates  $M$  and  $B$  is linearly independent. An  $R$ -module that has a basis is called a free module.

## Lecture 7 — Free Modules and Chain Conditions

**Theorem 30** (5.19). Let  $M$  be a free  $R$ -module with basis  $\{f_1, \dots, f_n\}$ . Then  $M \cong R^n$  as  $R$ -modules.

*Proof.* Define  $\varphi : M \rightarrow R^n$  by  $\varphi(\sum_{i=1}^n r_i f_i) = (r_1, \dots, r_n)$ , i.e. read off the coordinates in the basis. Since every element of  $M$  has a unique representation as  $\sum r_i f_i$  (by linear independence of the  $f_i$ ),  $\varphi$  is well-defined. It is clearly an  $R$ -homomorphism, injective (if  $(r_1, \dots, r_n) = 0$  then each  $r_i = 0$ , so  $m = 0$ ), and surjective (every  $(r_1, \dots, r_n) \in R^n$  is the image of  $\sum r_i f_i$ ).  $\square$

**Theorem 31** (5.20). *Let  $M$  be a finitely generated free  $R$ -module. Then every basis of  $M$  is finite.*

*Proof.* Let  $\{e_i\}_{i \in \Lambda}$  be a basis and  $\{x_1, \dots, x_n\}$  a finite generating set. Since elements of a module are finite sums by definition, each  $x_j$  involves only finitely many nonzero coefficients, so is supported on a finite subset  $S_j \subseteq \Lambda$ . Every basis element must appear in the expression of some generator (otherwise it could not be reached), so  $\Lambda \subseteq S_1 \cup \dots \cup S_n$ , a finite union of finite sets. Hence  $\Lambda$  is finite.  $\square$

**Theorem 32** (5.21). *Let  $M$  be a finitely generated free module over a commutative ring  $R$ . Then all bases of  $M$  have the same number of elements.*

*Proof.* Suppose  $M$  has two bases of sizes  $m$  and  $n$  respectively. By Theorem 5.19,  $M \cong R^m$  and  $M \cong R^n$ , so  $R^m \cong R^n$ . It suffices to show  $m = n$ .

Assume for contradiction  $m < n$ . Let  $\phi : R^m \rightarrow R^n$  be an  $R$ -isomorphism with inverse  $\psi = \phi^{-1}$ . Let  $(e_1, \dots, e_m)$  and  $(f_1, \dots, f_n)$  be the standard bases of  $R^m$  and  $R^n$ . Since  $\phi$  is an  $R$ -homomorphism, it is determined by where it sends each  $e_i$ , and each image lands in  $R^n$ , so we can write

$$\phi(e_i) = \sum_{j=1}^n a_{ji} f_j, \quad \psi(f_j) = \sum_{k=1}^m b_{kj} e_k.$$

Let  $A = (a_{ji})$  be the resulting  $n \times m$  matrix and  $B = (b_{kj})$  the  $m \times n$  matrix. Since  $\psi\phi = \text{id}_{R^m}$  and  $\phi\psi = \text{id}_{R^n}$ , substituting the expressions above and using linear independence of the bases gives  $BA = I_m$  and  $AB = I_n$ .

Pad these to  $n \times n$  matrices by setting  $A' = \begin{pmatrix} A & 0 \end{pmatrix}$  and  $B' = \begin{pmatrix} B \\ 0 \end{pmatrix}$ . Then  $A'B' = I_n$ , so  $\det(A'B') = 1$ . But  $B'A' = \begin{pmatrix} I_m & 0 \\ 0 & 0 \end{pmatrix}$ , so  $\det(B'A') = 0$ . Since  $R$  is commutative, determinants multiply, giving

$$1 = \det(A'B') = \det(A') \det(B') = \det(B'A') = 0,$$

a contradiction. By symmetry  $n < m$  also fails, so  $m = n$ .  $\square$

**Definition 13** (5.22 — Rank). *The number of elements in any basis of a finitely generated free module  $M$  over a commutative ring  $R$  is called the rank of  $M$ , written  $\text{rk}(M)$ . If  $R$  is a field, the rank is called the dimension.*

**Lemma 2** (5.24). *Every finitely generated  $R$ -module is a homomorphic image of a finitely generated free module.*

*Proof.* Let  $M$  be generated by  $x_1, \dots, x_n$ . Define  $\phi : R^n \rightarrow M$  by  $\phi(r_1, \dots, r_n) = \sum_{i=1}^n r_i x_i$ . This is a surjective  $R$ -homomorphism (surjective since the  $x_i$  generate  $M$ ). The generators  $x_i$  may satisfy relations, so  $\phi$  need not be injective — but by the First Isomorphism Theorem  $M \cong R^n / \ker \phi$ , expressing  $M$  as a quotient of the free module  $R^n$ .  $\square$

**Lemma 3** (5.25). *Let  $R$  be a PID and  $F$  a free  $R$ -module of rank  $n$ . Then every submodule  $K \leq F$  is free of rank  $m \leq n$ .*

*Proof.* By Theorem 5.19,  $F \cong R^n$ , so we work with  $K \leq R^n$  and induct on  $n$ . The base case  $n = 0$  is trivial since  $K = 0$  is free on the empty set.

Define  $\pi : R^n \rightarrow R$  by  $\pi(x_1, \dots, x_n) = x_1$ , and restrict it to  $K$ .

*Case 1:*  $\pi|_K = 0$ . Every element of  $K$  has first coordinate zero, so  $K \leq \{0\} \times R^{n-1} \cong R^{n-1}$ . By induction  $K$  is free of rank  $\leq n - 1 \leq n$ .

*Case 2:*  $\pi|_K \neq 0$ . The image  $\pi(K)$  is a nonzero ideal of  $R$ . Since  $R$  is a PID,  $\pi(K) = Ra$  for some  $a \neq 0$ . Pick  $k \in K$  with  $\pi(k) = a$ .

We claim  $K = Rk \oplus \ker(\pi|_K)$ , where  $\ker(\pi|_K) = \{x \in K \mid x_1 = 0\}$  is the set of elements of  $K$  with first coordinate zero, so  $\ker(\pi|_K) \leq \{0\} \times R^{n-1} \cong R^{n-1}$ .

$K = Rk + \ker(\pi|_K)$ : For any  $x \in K$ , write  $\pi(x) = ba$  for some  $b \in R$ . Then  $\pi(x - bk) = ba - ba = 0$ , so  $x - bk \in \ker(\pi|_K)$ , and  $x = bk + (x - bk)$ .

*The sum is direct:* If  $ck \in \ker(\pi|_K)$ , then  $0 = \pi(ck) = ca$ . Since  $R$  is a PID (hence an integral domain) and  $a \neq 0$ , we get  $c = 0$ .

By induction  $\ker(\pi|_K)$  is free of rank  $s \leq n - 1$ , and  $Rk \cong R$  is free of rank 1. Hence  $K \cong R^s \oplus R = R^{s+1}$ , which has rank  $m = s + 1 \leq n$ .  $\square$

## Lecture 8 — Noetherian and Artinian Modules

**Definition 14** (6.1 — Noetherian and Artinian modules). *An  $R$ -module  $M$  is called Noetherian if every ascending chain of submodules*

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \dots$$

*stabilises, i.e. there exists  $k$  such that  $M_k = M_{k+1} = \dots$ . This is called the ascending chain condition (ACC). An  $R$ -module  $M$  is called Artinian if every descending chain of submodules stabilises (descending chain condition, DCC).*

**Theorem 33** (6.5). *Let  $M$  be an  $R$ -module. The following are equivalent:*

- (i)  $M$  is Noetherian;
- (ii) every submodule of  $M$  is finitely generated;
- (iii) every non-empty family  $S$  of submodules of  $M$  has a maximal element.

*Proof.* (i)  $\Rightarrow$  (ii): Let  $N \leq M$  and suppose  $N$  is not finitely generated. For any finite choice  $a_1, \dots, a_k \in N$  we have  $N \neq (a_1, \dots, a_k)$ , so we can always pick  $a_{k+1} \in N \setminus (a_1, \dots, a_k)$ . This produces an infinite strictly ascending chain  $(a_1) \subsetneq (a_1, a_2) \subsetneq \dots$ , contradicting that  $M$  is Noetherian.

(ii)  $\Rightarrow$  (iii): Suppose  $S$  has no maximal element. Then starting from any  $N_0 \in S$  we can always find a strictly larger member, producing an infinite strictly ascending chain  $N_0 \subsetneq N_1 \subsetneq N_2 \subsetneq \dots$ . Let  $N = \bigcup_i N_i$ . For  $x, y \in N$  and  $r \in R$ , both  $x$  and  $y$  lie in some  $N_\ell$  (take the larger index), so  $x - y, rx \in N_\ell \subseteq N$ ; hence  $N$  is a submodule. By (ii),  $N = (a_1, \dots, a_n)$  for some  $a_j \in N$ . Each  $a_j$  lies in some  $N_{u_j}$ , so all generators lie in  $N_\mu$  for  $\mu = \max\{u_1, \dots, u_n\}$ . But then  $N = (a_1, \dots, a_n) \subseteq N_\mu \subseteq N$ , so  $N_\mu = N$ , contradicting that the chain is strictly ascending.

(iii)  $\Rightarrow$  (i): Given an ascending chain  $M_1 \subseteq M_2 \subseteq \dots$ , by (iii) the family  $\{M_i\}$  has a maximal element  $M_k$ . Then  $M_k = M_{k+1} = \dots$ , so the chain stabilises.  $\square$

**Definition 15** (6.6 — Finitely cogenerated). An  $R$ -module  $M$  is finitely cogenerated if for every family  $(M_\alpha)_{\alpha \in \Lambda}$  of submodules of  $M$ ,

$$\bigcap_{\alpha \in \Lambda} M_\alpha = 0 \implies \bigcap_{\alpha \in \Lambda'} M_\alpha = 0$$

for some finite subset  $\Lambda' \subseteq \Lambda$ .

**Theorem 34** (6.8). Let  $M$  be an  $R$ -module. The following are equivalent:

- (i)  $M$  is Artinian;
- (ii) every quotient module of  $M$  is finitely cogenerated;
- (iii) every non-empty family  $S$  of submodules of  $M$  has a minimal element.

*Proof.* (i)  $\Rightarrow$  (ii): Let  $N \leq M$  and take any family  $(M_\alpha/N)_{\alpha \in \Lambda}$  of submodules of  $M/N$  with  $\bigcap_{\alpha \in \Lambda} M_\alpha/N = 0$ , i.e.  $\bigcap_{\alpha \in \Lambda} M_\alpha = N$ . Consider the family of finite intersections  $\{\bigcap_{\alpha \in \Lambda'} M_\alpha \mid \Lambda' \subseteq \Lambda \text{ finite}\}$ . As we add more submodules to the intersection it can only get smaller, so this is a descending family of submodules of  $M$ . Since  $M$  is Artinian every descending chain stabilises, so there exists a finite  $\Lambda' \subseteq \Lambda$  such that

$$\bigcap_{\alpha \in \Lambda'} M_\alpha = \bigcap_{\alpha \in \Lambda} M_\alpha = N.$$

Hence  $\bigcap_{\alpha \in \Lambda'} M_\alpha/N = 0$ , so  $M/N$  is finitely cogenerated.

(ii)  $\Rightarrow$  (iii): Suppose  $S$  has no minimal element. Then starting from any  $N_0 \in S$  we can always find a strictly smaller member, producing an infinite strictly descending chain  $N_0 \supsetneq N_1 \supsetneq N_2 \supsetneq \dots$ . Let  $N = \bigcap_i N_i$ , which is a submodule of  $M$ . By (ii),  $M/N$  is finitely cogenerated. The family  $(N_i/N)$  of submodules of  $M/N$  satisfies  $\bigcap_i N_i/N = N/N = 0$ , so by finite cogeneratedness there exists a finite subset, say  $\{N_0, \dots, N_k\}$ , with  $\bigcap_{i=0}^k N_i/N = 0$ , i.e.  $\bigcap_{i=0}^k N_i = N$ . Since the chain is descending,  $\bigcap_{i=0}^k N_i = N_k$ , so  $N_k = N \subseteq N_{k+1} \subsetneq N_k$ , a contradiction.

(iii)  $\Rightarrow$  (i): Given a descending chain  $M_1 \supseteq M_2 \supseteq \dots$ , by (iii) the family  $\{M_i\}$  has a minimal element  $M_k$ . Then  $M_k = M_{k+1} = \dots$ , so the chain stabilises.  $\square$

**Theorem 35** (6.9). Let  $M$  be a Noetherian (respectively Artinian)  $R$ -module and  $K \leq M$  a submodule. Then  $K$  and  $M/K$  are both Noetherian (respectively Artinian).

*Proof. Noetherian case.* By Theorem 6.5,  $M$  is Noetherian if and only if every submodule of  $M$  is finitely generated. Every submodule of  $K$  is also a submodule of  $M$ , hence finitely generated. By Theorem 6.5 again,  $K$  is Noetherian. For  $M/K$ : by Theorem 4.31 (Correspondence Theorem), every submodule of  $M/K$  is of the form  $L/K$  for some submodule  $L$  with  $K \subseteq L \subseteq M$ . Since  $L$  is a submodule of  $M$ , it is finitely generated, and hence  $L/K$  is finitely generated. By Theorem 6.5,  $M/K$  is Noetherian.

*Artinian case.* Any descending chain  $K \supseteq L_1 \supseteq L_2 \supseteq \dots$  of submodules of  $K$  is also a descending chain in  $M$ , so it stabilises. Hence  $K$  is Artinian. For  $M/K$ : a descending chain  $M/K \supseteq L_1/K \supseteq L_2/K \supseteq \dots$  of submodules of  $M/K$  corresponds via Theorem 4.31 to a descending chain  $M \supseteq L_1 \supseteq L_2 \supseteq \dots$  in  $M$ , which stabilises. Hence  $M/K$  is Artinian.  $\square$

**Theorem 36** (6.10). *Let  $M$  be an  $R$ -module and  $N \leq M$  a submodule. Then  $M$  is Noetherian (respectively Artinian) if and only if both  $N$  and  $M/N$  are Noetherian (respectively Artinian).*

*Proof.* ( $\Rightarrow$ ) This is Theorem 6.9.

( $\Leftarrow$ ) *Noetherian case.* Assume  $N$  and  $M/N$  are Noetherian. Let  $K \leq M$  be any submodule; we show  $K$  is finitely generated, which by Theorem 6.5 gives that  $M$  is Noetherian.

By the Second Isomorphism Theorem,  $K/(N \cap K) \cong (K + N)/N$ . Since  $(K + N)/N$  is a submodule of  $M/N$  and  $M/N$  is Noetherian, it is finitely generated by Theorem 6.5, and hence so is  $K/(N \cap K)$ . Say  $K/(N \cap K)$  is generated by  $\bar{x}_1, \dots, \bar{x}_m$  with  $x_i \in K$ . Then

$$K = (x_1, \dots, x_m) + (N \cap K).$$

Since  $N \cap K$  is a submodule of  $N$  and  $N$  is Noetherian,  $N \cap K$  is finitely generated by Theorem 6.5, say by  $y_1, \dots, y_n \in N \cap K$ . Therefore

$$K = (x_1, \dots, x_m, y_1, \dots, y_n),$$

so  $K$  is finitely generated and  $M$  is Noetherian.

*Artinian case.* Assume  $N$  and  $M/N$  are Artinian. Let  $M \supseteq K_1 \supseteq K_2 \supseteq \dots$  be a descending chain in  $M$ . The induced chains

$$K_1 \cap N \supseteq K_2 \cap N \supseteq \dots \quad \text{in } N, \quad \frac{K_1 + N}{N} \supseteq \frac{K_2 + N}{N} \supseteq \dots \quad \text{in } M/N$$

both stabilise (since  $N$  and  $M/N$  are Artinian), say at indices  $n$  and  $m$  respectively. Set  $q = \max\{m, n\}$ . For  $i \geq q$  we have  $K_i \cap N = K_q \cap N$  and  $K_i + N = K_q + N$ . One can then show using the Second Isomorphism Theorem that  $K_i = K_q$ , so the original chain stabilises and  $M$  is Artinian.  $\square$

**Corollary 6** (6.11). *If  $M_1, \dots, M_n$  are Noetherian (respectively Artinian)  $R$ -modules, then  $M_1 \oplus \dots \oplus M_n$  is Noetherian (respectively Artinian).*

*Proof.* By induction it suffices to treat the case of two modules  $M$  and  $N$ . By Theorem 6.10, to show  $M \oplus N$  is Noetherian (respectively Artinian) it suffices to find a single submodule of  $M \oplus N$  such that both the submodule and its quotient module are Noetherian (respectively Artinian). Take  $M \oplus 0 \leq M \oplus N$ . Then:

- the submodule  $M \oplus 0 \cong M$  is Noetherian (respectively Artinian) by assumption;
- the quotient  $(M \oplus N)/(M \oplus 0) \cong N$  is Noetherian (respectively Artinian) by assumption.

Hence by Theorem 6.10,  $M \oplus N$  is Noetherian (respectively Artinian).  $\square$

**Definition 16** (6.12 — Noetherian and Artinian rings). *A ring  $R$  is called left Noetherian (respectively left Artinian) if  $R$ , regarded as a left  $R$ -module, is Noetherian (respectively Artinian). By default, Noetherian (respectively Artinian) ring means left Noetherian (respectively left Artinian).*

**Theorem 37** (6.13). *Let  $R$  be a ring. The following are equivalent:*



- (i)  $R$  is Noetherian (respectively Artinian);
- (ii) every left ideal  $A$  of  $R$  is finitely generated (respectively every  $R/A$  is finitely cogenerated);
- (iii) every non-empty set of left ideals of  $R$  has a maximal (respectively minimal) element.

*Proof.* This is Theorem 6.5 (respectively 6.8) applied to  $R$  regarded as a left  $R$ -module, where submodules of  $R$  are precisely the left ideals of  $R$ .  $\square$

**Theorem 38** (6.19). *Let  $R_1, \dots, R_n$  be Noetherian (respectively Artinian) rings. Then  $R_1 \oplus \dots \oplus R_n$  is Noetherian (respectively Artinian).*

*Proof.* This follows immediately from Corollary 6.11, viewing each  $R_i$  as an  $R$ -module.  $\square$

**Product of ideals.** For ideals  $A$  and  $B$  in a ring  $R$ , define

$$AB = \left\{ \sum_{i=1}^n a_i b_i \mid n \in \mathbb{N}, a_i \in A, b_i \in B \right\}.$$

Then  $AB$  is again an ideal of  $R$ .

**Definition 17** (6.20 — Prime ideal). *Let  $R$  be a ring. An ideal  $P \neq R$  is prime if whenever  $A$  and  $B$  are ideals in  $R$  with  $AB \subseteq P$ , then  $A \subseteq P$  or  $B \subseteq P$ .*

(For commutative  $R$  this recovers the classical definition:  $ab \in P \Rightarrow a \in P$  or  $b \in P$ .)

**Theorem 39** (6.21). *Let  $R$  be a ring with unity. Then every maximal ideal is prime.*

*Proof.* Let  $M$  be a maximal ideal and let  $A, B$  be ideals with  $AB \subseteq M$ . Suppose  $A \not\subseteq M$ . Since  $M$  is maximal, the only ideal strictly containing  $M$  is  $R$  itself, so  $A + M = R$ . Write  $1 = a + m$  for some  $a \in A$  and  $m \in M$ . For any  $b \in B$ ,

$$b = 1 \cdot b = (a + m)b = ab + mb \in AB + MB \subseteq M + M = M.$$

Hence  $B \subseteq M$ , so  $M$  is prime.  $\square$

*Note:* The converse is false:  $(0)$  is a prime ideal in  $\mathbb{Z}$  but not a maximal ideal.

## Lecture 9

**Lemma 4** (6.22). *If  $R$  is a Noetherian ring, then each ideal contains a finite product of prime ideals.*

*Proof.* Let  $\mathcal{F}$  be the family of ideals of  $R$  that do not contain any finite product of prime ideals. We want to show  $\mathcal{F} = \emptyset$ .

Suppose for a contradiction that  $\mathcal{F} \neq \emptyset$ . Since  $R$  is Noetherian, by Theorem 6.13(iii) the family  $\mathcal{F}$  has a maximal element, say  $A$ .

We claim  $A$  is not a prime ideal. Indeed, any prime ideal  $P$  contains the one-factor product  $P$  itself, so no prime ideal can belong to  $\mathcal{F}$ .

Since  $A$  is not prime, there exist ideals  $B$  and  $C$  with  $BC \subseteq A$ , but  $B \not\subseteq A$  and  $C \not\subseteq A$ . Then  $B + A \supsetneq A$  and  $C + A \supsetneq A$ , so by maximality of  $A$  in  $\mathcal{F}$ , neither  $B + A$  nor  $C + A$  belongs to  $\mathcal{F}$ . Hence both contain finite products of prime ideals. But then  $(B + A)(C + A)$  also contains a finite product of prime ideals, and

$$(B + A)(C + A) \subseteq BC + BA + AC + A^2 \subseteq A.$$

So  $A$  contains a finite product of prime ideals, contradicting  $A \in \mathcal{F}$ . Hence  $\mathcal{F} = \emptyset$ .  $\square$

**Theorem 40** (6.23 — Hilbert's Basis Theorem). *Let  $R$  be a Noetherian ring. Then the polynomial ring  $R[x]$  is also Noetherian.*

*Proof.* Suppose for a contradiction that  $I \subseteq R[x]$  is a left ideal that is not finitely generated. We recursively construct a sequence of polynomials as follows. Set

$$J_0 = (0) \subseteq R[x].$$

Since  $I$  is not finitely generated,  $I \setminus J_0 \neq \emptyset$ . Pick  $f_0 \in I \setminus J_0$  of minimal degree (exists since degrees are non-negative integers). For  $n \geq 1$ , given  $f_0, \dots, f_{n-1}$ , let

$$J_n = (f_0, \dots, f_{n-1}) \subseteq R[x]$$

be the left ideal in  $R[x]$  generated by  $f_0, \dots, f_{n-1}$ . Since  $I$  is not finitely generated,  $I \setminus J_n \neq \emptyset$ , so we may pick  $f_n \in I \setminus J_n$  of minimal degree.

Since  $J_n \subsetneq J_{n+1}$ , the set  $I \setminus J_{n+1} \subsetneq I \setminus J_n$ , so  $\deg(f_n)$  is non-decreasing in  $n$ .

For each  $n$ , let  $a_n \in R$  denote the leading coefficient of  $f_n$ . Consider the ascending chain of left ideals in  $R$ :

$$L_1 \subseteq L_2 \subseteq \dots, \quad \text{where } L_n = (a_0, \dots, a_{n-1}) \subseteq R.$$

Since  $R$  is Noetherian, this chain stabilises: there exists  $N \in \mathbb{N}$  such that  $L_N = L_{N+1}$ , that is,

$$a_N \in L_N = (a_0, \dots, a_{N-1}) \subseteq R.$$

So we may write  $a_N = \sum_{i < N} u_i a_i$  for some  $u_i \in R$ . Now define

$$g = \sum_{i < N} u_i x^{\deg(f_N) - \deg(f_i)} f_i \in J_N.$$

The exponents  $\deg(f_N) - \deg(f_i) \geq 0$  since degrees are non-decreasing. The leading term of  $u_i x^{\deg(f_N) - \deg(f_i)} f_i$  is  $u_i a_i x^{\deg(f_N)}$ , so the leading term of  $g$  is

$$\left( \sum_{i < N} u_i a_i \right) x^{\deg(f_N)} = a_N x^{\deg(f_N)},$$

which equals the leading term of  $f_N$ . Hence  $f_N - g$  has degree strictly less than  $\deg(f_N)$ .

Since  $f_N \notin J_N$  and  $g \in J_N$ , we have  $f_N - g \notin J_N$ . Since  $f_N, g \in I$ , we have  $f_N - g \in I$ . Thus  $f_N - g \in I \setminus J_N$ .

- If  $\deg(f_N) > 0$ : then  $f_N - g \in I \setminus J_N$  has degree strictly less than  $\deg(f_N)$ , contradicting the minimality of  $\deg(f_N)$ .

- If  $\deg(f_N) = 0$ : then all degrees are 0, so  $f_N - g = 0$ , meaning  $f_N = g \in J_N$ , contradicting  $f_N \in I \setminus J_N$ .

In both cases we reach a contradiction, so  $I$  must be finitely generated and  $R[x]$  is Noetherian.  $\square$

**Definition 18** (7.1 — Nilpotent elements and ideals). *Let  $R$  be a ring.*

- An element  $a \in R$  is nilpotent if  $a^n = 0$  for some  $n \in \mathbb{N}$ .
- A left ideal  $I$  of  $R$  is a nil ideal if every element of  $I$  is nilpotent.
- A left ideal  $I$  of  $R$  is nilpotent if  $I^n = 0$  for some  $n \in \mathbb{N}$ , where  $I^n$  denotes the set of all finite sums of products  $i_1 i_2 \cdots i_n$  with each  $i_k \in I$ .

Every nilpotent ideal is nil (if  $I^n = 0$  then any  $a \in I$  satisfies  $a^n \in I^n = 0$ ), but the converse fails in general.

**Theorem 41** (7.5). *If  $J$  is a nil left ideal in an Artinian ring  $R$ , then  $J$  is nilpotent.*

*Proof.* Suppose for a contradiction that  $J^k \neq 0$  for all  $k \in \mathbb{N}$ . The family  $(J, J^2, J^3, \dots)$  is a descending chain of left ideals of  $R$ . Since  $R$  is Artinian, this family has a minimal element, say  $B = J^m$  for some  $m$ . Then

$$B^2 = J^{2m} \subseteq J^m = B,$$

and by minimality of  $B$  we have  $B^2 = B$ .

Now consider the family

$$\mathcal{F} = \{A \mid A \text{ is a left ideal of } R, A \subseteq B, BA \neq 0\}.$$

Since  $B^2 = B \neq 0$ , we have  $B \in \mathcal{F}$ , so  $\mathcal{F} \neq \emptyset$ . Since  $R$  is Artinian,  $\mathcal{F}$  has a minimal element  $A$ .

Since  $BA \neq 0$ , there exists  $a \in A$  with  $Ba \neq 0$ . Note that  $Ba = \{ba \mid b \in B\}$  is a left ideal contained in  $A$ , and

$$B(Ba) = B^2 a = Ba \neq 0,$$

so  $Ba \in \mathcal{F}$ . By minimality of  $A$  we conclude  $Ba = A$ .

Hence there exists  $b \in B$  with  $ba = a$ . Applying  $b$  repeatedly gives  $b^i a = a$  for all  $i \in \mathbb{N}$ . But  $b \in B \subseteq J$  and  $J$  is nil, so  $b^n = 0$  for some  $n \in \mathbb{N}$ . Then  $a = b^n a = 0$ , contradicting  $Ba \neq 0$ .

Therefore  $J^k = 0$  for some  $k \in \mathbb{N}$ , i.e.  $J$  is nilpotent.  $\square$

**Lemma 5** (7.6). *Let  $R$  be a Noetherian ring. Then the sum of nilpotent left ideals in  $R$  is a nilpotent left ideal.*

*Proof.* Let  $B = \sum_{i \in \Lambda} A_i$  be a sum of nilpotent left ideals. Since  $R$  is Noetherian,  $B$  is finitely generated as a left ideal, say  $B = (x_1, \dots, x_m)$ . Each  $x_j$  lies in a finite sum of the  $A_i$ 's, so  $B$  is contained in a finite sum  $A_1 + \cdots + A_n$  (reindexing if necessary), and hence  $B = A_1 + \cdots + A_n$ .

It remains to show  $B$  is nilpotent. For each  $i$ , let  $n_i \in \mathbb{N}$  be such that  $A_i^{n_i} = 0$ . Set  $N = n_1 + \cdots + n_n$ . Any product of  $N$  elements from  $B = A_1 + \cdots + A_n$ , when expanded, is a sum of products of the form  $a_{i_1} a_{i_2} \cdots a_{i_N}$  with each  $a_{i_k} \in A_{i_k}$ . By pigeonhole, among the  $N$  indices  $i_1, \dots, i_N \in \{1, \dots, n\}$ , at least one value  $j$  appears at least  $n_j$  times. The consecutive factors from  $A_j$  multiply to give an element of  $A_j^{n_j} = 0$ . Hence every such product is 0, so  $B^N = 0$ .  $\square$

## Lecture 10

**Theorem 42** (7.7). *Let  $R$  be a Noetherian ring having no non-zero nilpotent ideals. Then  $R$  has no non-zero nil ideals.*

(Proof non-examinable.)

**Theorem 43** (7.8). *Let  $N$  be a nil ideal in a Noetherian ring  $R$ . Then  $N$  is nilpotent.*

*Proof.* Let  $T$  be the sum of all nilpotent ideals in  $R$ . By Lemma 7.6,  $T$  is itself nilpotent, say  $T^k = 0$ .

We claim  $R/T$  has no non-zero nilpotent ideals. Suppose  $A/T$  is a nilpotent ideal in  $R/T$ , say  $(A/T)^m = 0$ . Then  $(A^m + T)/T = 0$ , so  $A^m \subseteq T$ . Since  $T^k = 0$ , we get  $A^{mk} \subseteq T^k = 0$ , so  $A$  is nilpotent. Hence  $A \subseteq T$ , giving  $A/T = 0$ .

Since  $R/T$  is Noetherian (a quotient of a Noetherian ring) and has no non-zero nilpotent ideals, Theorem 7.7 gives that  $R/T$  has no non-zero nil ideals. Now  $(N + T)/T$  is a nil ideal in  $R/T$  (it is nil since  $N$  is nil and  $T$  is nilpotent hence nil), so  $(N + T)/T = 0$ , which means  $N \subseteq T$ . Since  $T$  is nilpotent,  $N$  is nilpotent.  $\square$

**Lemma 6** (7.9). *Let  $A$  be a minimal left ideal in a ring  $R$ . Then either  $A^2 = 0$  or  $A = Re$  for some idempotent  $e \in R$ .*

*Proof.* Assume  $A^2 \neq 0$ .

*Step 1: Find  $e \in A$  with  $ea = a$ .* Since  $A^2 \neq 0$ , there exists  $a \in A$  with  $Aa \neq 0$ . The set  $Aa = \{ca \mid c \in A\}$  is a left ideal contained in  $A$  (since  $A$  is a left ideal and  $a \in A$ ). By minimality of  $A$ , we have  $Aa = A$ . In particular, there exists  $e \in A$  with  $ea = a$ .

*Step 2: Show  $e^2 = e$ .* We compute  $(e^2 - e)a = e(ea) - ea = ea - ea = 0$ . To conclude  $e^2 = e$ , define the left annihilator of  $a$  in  $A$ :

$$\text{Ann}_A(a) := \{c \in A \mid ca = 0\}.$$

This is a left ideal of  $R$  contained in  $A$ . Since  $ea = a \neq 0$ , we have  $e \notin \text{Ann}_A(a)$ , so  $\text{Ann}_A(a) \neq A$ . By minimality of  $A$ , we conclude  $\text{Ann}_A(a) = 0$ . Since  $e^2 - e \in A$  and  $(e^2 - e)a = 0$ , we get  $e^2 - e \in \text{Ann}_A(a) = 0$ , hence  $e^2 = e$ .

*Step 3: Show  $A = Re$ .* Since  $e \in A$  and  $A$  is a left ideal,  $Re \subseteq A$ . Since  $e \neq 0$  (as  $ea = a \neq 0$ ), we have  $Re \neq 0$ . By minimality of  $A$ ,  $Re = A$ .  $\square$

**Theorem 44** (7.10 — Wedderburn–Artin). *Let  $R$  be a left (or right) Artinian ring with unity and no non-zero nilpotent ideals. Then*

$$R \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k)$$

for some  $n_1, \dots, n_k \in \mathbb{N}$  and division rings  $D_1, \dots, D_k$ .

*Proof.* We build the result from two claims, then combine them.

**Claim 1.** *Each non-zero left ideal  $A$  of  $R$  is of the form  $Re$  for some idempotent  $e \in R$ .*

By Lemma 7.9, every minimal left ideal is either nilpotent or of the form  $Re$  for some idempotent  $e$ . Since  $R$  has no non-zero nilpotent ideals, every minimal left ideal is of the form  $Re$ .

Now let  $A$  be any non-zero left ideal. Since  $R$  is Artinian,  $A$  contains a minimal left ideal, hence a non-zero idempotent. Thus the family

$$\mathcal{F} = \{R(1 - f) \cap A \mid f \text{ a non-zero idempotent in } A\}$$

is non-empty, and by the Artinian condition has a minimal member  $R(1 - e) \cap A$ .

**Subclaim** (*non-examinable*).  $R(1 - e) \cap A = 0$ .

Invoking the subclaim: for any  $a \in A$ ,  $a(1 - e) \in R(1 - e) \cap A = 0$ , so  $a = ae \in Re$ . Hence  $A \subseteq Re \subseteq A$ , giving  $A = Re$ .

**Claim 2.**  $R = Re_1 \oplus \cdots \oplus Re_n$  is a finite direct sum of minimal left ideals.

Let  $S$  be the sum of all minimal left ideals in  $R$ ; since  $R$  is Artinian it has at least one minimal left ideal, so  $S \neq 0$ . By Claim 1,  $S = Re$  for some idempotent  $e$ .

Suppose  $R(1 - e) \neq 0$ . Then  $R(1 - e)$  is a non-zero left ideal, so by the Artinian condition it contains a minimal left ideal  $A$ . But  $A \subseteq S = Re$ , giving  $A \subseteq Re \cap R(1 - e)$ . For any  $x \in Re \cap R(1 - e)$ , write  $x = ae = b(1 - e)$  for some  $a, b \in R$ . Then  $xe = ae^2 = ae = x$ , and also  $xe = b(1 - e)e = b(e - e^2) = 0$ , so  $x = 0$ . Hence  $Re \cap R(1 - e) = 0$ , giving  $A = 0$ , a contradiction.

Therefore  $R(1 - e) = 0$ . Since  $R$  has unity,  $1 - e = 0$ , i.e.  $e = 1$ , so  $S = R$ .

Thus  $R$  is the sum of all its minimal left ideals. By Theorem 5.9 there exists a subfamily  $(A_i)_{i \in \Lambda'}$  such that  $R = \bigoplus_{i \in \Lambda'} A_i$ . To see this is finite, write  $1 = e_{i_1} + \cdots + e_{i_n}$  as a finite sum with  $e_{i_j} \in A_{i_j}$ . Then for any  $r \in R$ ,  $r = r \cdot 1 = re_{i_1} + \cdots + re_{i_n}$ , so  $R = Re_{i_1} \oplus \cdots \oplus Re_{i_n}$ . After reindexing,  $R = Re_1 \oplus \cdots \oplus Re_n$ .

**Conclusion.** From Claim 2,  $R = Re_1 \oplus \cdots \oplus Re_n$ . Among  $Re_1, \dots, Re_n$ , select a maximal pairwise non-isomorphic subfamily; after reindexing call it  $N_1 = Re_1, \dots, N_k = Re_k$ . For each  $j$ , let  $n_j$  be the number of summands isomorphic to  $N_j$ , so that

$$R = \underbrace{(N_1 \oplus \cdots \oplus N_1)}_{n_1} \oplus \cdots \oplus \underbrace{(N_k \oplus \cdots \oplus N_k)}_{n_k}.$$

By Schur's Lemma 5.6, any non-zero homomorphism between simple modules is an isomorphism. Therefore  $\text{Hom}_R(N_i, N_j) = 0$  whenever  $i \neq j$  (since  $N_i \not\cong N_j$ ), and  $D_j := \text{Hom}_R(N_j, N_j)$  is a division ring. Invoking Theorem 4.30 on the decomposition above, the off-diagonal blocks vanish and we get

$$\text{Hom}_R(R, R) \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k).$$

It remains to identify  $\text{Hom}_R(R, R)$  with  $R$ . Every  $\varphi \in \text{Hom}_R(R, R)$  is determined by  $b := \varphi(1)$ , since  $\varphi(r) = \varphi(r \cdot 1) = r\varphi(1) = rb$ . So every endomorphism is right multiplication  $\rho_b$  by some  $b \in R$ . However composition reverses order:  $(\rho_b \circ \rho_c)(1) = \rho_b(c) = cb$ , which is the product  $b \cdot_{\text{op}} c$  in  $R^{\text{op}}$ . Hence  $\text{Hom}_R(R, R) \cong R^{\text{op}}$  as rings, giving

$$R^{\text{op}} \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k).$$

Passing to opposite rings: the opposite of a direct sum is the direct sum of the opposites,  $(M_n(D))^{\text{op}} \cong M_n(D^{\text{op}})$ , and  $D^{\text{op}}$  is again a division ring. Relabelling  $D_j^{\text{op}}$  as  $D_j$  gives  $R \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k)$ .  $\square$

**Theorem 45** (7.11). *Let  $R$  be a left (or right) Artinian ring with unity and no non-zero nilpotent ideals. Then  $R$  is also right and left Artinian, and right and left Noetherian.*

(Non-examinable. Follows from Wedderburn–Artin and the fact that matrix rings over division rings are Noetherian and Artinian on both sides.)

**Theorem 46** (7.13 — Maschke). *If  $G$  is a finite group, then*

$$\mathbb{C}[G] \cong M_{n_1}(\mathbb{C}) \oplus \cdots \oplus M_{n_k}(\mathbb{C})$$

*for some  $n_1, \dots, n_k \in \mathbb{N}$ .*

(Non-examinable. Follows from Wedderburn–Artin: one shows  $\mathbb{C}[G]$  has no non-zero nilpotent ideals and is Artinian, then uses that the only division ring containing  $\mathbb{C}$  in its centre as an algebraically closed field is  $\mathbb{C}$  itself.)

**Theorem 47** (7.14). *Let  $R$  be a ring with unity. The following are equivalent:*

- (i)  *$R$  is left Artinian with no non-zero nilpotent ideals;*
- (ii)  *$R$  is left Artinian with no non-zero nil ideals;*
- (iii)  *$R$  is a finite direct sum of minimal left ideals;*
- (iv) *every left ideal of  $R$  is of the form  $Re$  for some idempotent  $e$ ;*
- (v)  *$R$  is a finite direct sum of matrix rings over division rings.*

(Non-examinable.)

**Definition 19** (7.15 — Semisimple Artinian ring). *A ring satisfying any of the equivalent conditions of Theorem 7.14 is called a semisimple Artinian ring.*

## Lecture 11 — Smith Normal Form over a PID

Throughout this chapter  $R$  is a principal ideal domain with unity.

**Definition 20** (8.1 — Row module, column module). *Let  $A \in M_{m \times n}(R)$ . The submodule of  $R^n$  generated by the  $m$  rows of  $A$  is called the row module of  $A$ , denoted  $\mathcal{R}(A)$ . The submodule of  $R^m$  generated by the  $n$  columns of  $A$  is called the column module of  $A$ , denoted  $\mathcal{C}(A)$ .*

*Since  $\mathcal{R}(A)$  and  $\mathcal{C}(A)$  are finitely generated submodules of the free modules  $R^n$  and  $R^m$  respectively, both are free by Lemma 5.25.*

**Definition 21** (8.2 — Row rank, column rank). *Let  $A \in M_{m \times n}(R)$ . The rank of  $\mathcal{R}(A)$  is called the row rank of  $A$ , and the rank of  $\mathcal{C}(A)$  is called the column rank of  $A$ .*

**Theorem 48** (8.3). *Let  $A \in M_{m \times n}(R)$  and let  $P, Q$  be invertible  $m \times m$  and  $n \times n$  matrices over  $R$  respectively. Then row rank and column rank of  $PAQ$  equal those of  $A$ .*

(Proof non-examinable.)

**Definition 22** (8.4/8.6 — Elementary operations and matrices). *Let  $A \in M_{m \times n}(R)$ . There are three elementary row operations, each achieved by left-multiplying  $A$  by an invertible elementary matrix (column operations are achieved by right-multiplying):*

- (i) Swap rows  $i$  and  $j$ : multiply on the left by  $S_{ij} = I - E_{ii} - E_{jj} + E_{ij} + E_{ji}$ . Here  $S_{ij}^{-1} = S_{ij}$ .
- (ii) Scale row  $i$  by invertible  $\alpha \in R$ : multiply on the left by  $M_i(\alpha) = I + (\alpha - 1)E_{ii}$ . Here  $M_i(\alpha)^{-1} = M_i(\alpha^{-1})$ .
- (iii) Add  $\alpha \in R$  times row  $j$  to row  $i$ : multiply on the left by  $C_{ij}(\alpha) = I + \alpha E_{ij}$ . Here  $C_{ij}(\alpha)^{-1} = C_{ij}(-\alpha)$ .

Since each elementary matrix is invertible, any sequence of elementary row and column operations transforms  $A$  into an equivalent matrix  $PAQ$ .

**Definition 23** (8.7 — Matrix equivalence). Two matrices  $A, B \in M_{m \times n}(R)$  are equivalent if there exist invertible matrices  $P \in M_m(R)$  and  $Q \in M_n(R)$  such that  $B = PAQ$ .

**Theorem 49** (8.8 — Smith Normal Form). Let  $A \in M_{m \times n}(R)$  where  $R$  is a PID. Then  $A$  is equivalent to a matrix of the form

$$\begin{pmatrix} a_1 & & & \\ & \ddots & & \\ & & a_r & \\ & & & 0 \end{pmatrix},$$

where  $a_1, \dots, a_r \neq 0$  and  $a_1 \mid a_2 \mid \dots \mid a_r$ .

*Proof.* Define the length  $\ell(a)$  of a non-zero  $a \in R$  to be the number of prime factors in the factorisation  $a = p_1 \cdots p_k$  counted with multiplicity, with the convention  $\ell(u) = 0$  for units.

*Step 1: bring a minimal-length entry to position  $(1, 1)$ .* If  $A = 0$  the result is immediate. Otherwise, among all non-zero entries pick  $a_{ij}$  with minimal  $\ell(a_{ij})$ . Row and column swaps bring it to position  $(1, 1)$ .

*Step 2: clear the first row.* Fix any column  $k \geq 2$ . If  $a_{11} \mid a_{1k}$ , write  $a_{1k} = a_{11}q$  and apply  $C_k - qC_1$ , zeroing the  $(1, k)$  entry without changing  $a_{11}$ . If  $a_{11} \nmid a_{1k}$ , let  $d = \gcd(a_{11}, a_{1k})$  and write  $a_{11} = ds$ ,  $a_{1k} = dt$ . By Bézout there exist  $u, v \in R$  with  $us + vt = 1$ , so the matrix  $\begin{pmatrix} u & t \\ v & -s \end{pmatrix}$  is invertible (determinant  $-1$ ). Right-multiplying  $A$  by the  $n \times n$  matrix with this  $2 \times 2$  block in the  $(1, k)$ -positions and the identity elsewhere replaces the  $(1, 1)$  entry by  $a_{11}u + a_{1k}v = d$  and the  $(1, k)$  entry by  $a_{11}t - a_{1k}s = 0$ . Since  $d$  properly divides  $a_{11}$ , we have  $\ell(d) < \ell(a_{11})$ . Repeating across all columns clears the first row.

*Step 3: clear the first column.* The identical argument applied to rows (left-multiplying by the analogous block matrices) clears all entries below  $(1, 1)$ .

*Step 4: termination.* Clearing the first column may reintroduce non-zero entries in the first row, requiring another pass of Step 2. However this only occurs when  $a_{11}$  does not divide some first-column entry, which triggers the Bézout case and strictly reduces  $\ell(a_{11})$ . Since  $\ell(a_{11}) \geq 0$  is a non-negative integer, the alternating process terminates after at most  $\ell(a_{11})$  restarts, yielding

$$P_1 A Q_1 = \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A_1 & \\ 0 & & & \end{pmatrix},$$

where  $A_1$  is an  $(m-1) \times (n-1)$  submatrix.

*Step 5: induction.* Apply the same procedure to  $A_1$  and continue. By induction on  $m+n$  we obtain  $PAQ = \text{diag}(a_1, \dots, a_r, 0, \dots, 0)$ .

*Step 6: enforce  $a_1 \mid a_2 \mid \dots \mid a_r$ .* Suppose  $a_1 \nmid a_2$ . Add row 2 to row 1; the first row becomes  $(a_1, a_2, 0, \dots, 0)$ . Applying Step 2 replaces  $a_1$  by  $\gcd(a_1, a_2)$ , strictly reducing  $\ell(a_1)$ . Repeating forces  $a_1 \mid a_2$ , and similarly  $a_1 \mid a_i$  for all  $i$ . Repeating the argument for  $a_2$  in place of  $a_1$ , and so on, gives  $a_i \mid a_{i+1}$  for all  $i = 1, \dots, r-1$ .  $\square$

**Definition 24** (8.9 — Invariant factors). *The non-zero diagonal entries  $a_1, \dots, a_r$  of the Smith normal form of  $A$  are called the invariant factors of  $A$ . They are unique up to unit multipliers, and two  $m \times n$  matrices are equivalent if and only if they have the same invariant factors.*

**Theorem 50** (8.10). *Let  $A$  be an  $m \times n$  matrix over a PID. Then the row rank of  $A$  equals the column rank of  $A$ .*

*Proof.* Choose invertible  $P, Q$  such that  $PAQ$  is in Smith normal form (Theorem 8.8). By Theorem 8.3, multiplying by invertible matrices preserves row and column rank, so

$$\text{row rank}(A) = \text{row rank}(PAQ) = r = \text{col rank}(PAQ) = \text{col rank}(A),$$

where  $r$  is the number of non-zero diagonal entries.  $\square$

**Definition 25** (8.11 — Rank). *The common value of the row rank and column rank of  $A$  is called the rank of  $A$ , denoted  $\text{rank}(A)$ .*

## Lecture 12 — Finitely Generated Modules over a PID

**Theorem 51** (9.1 — Decomposition Theorem). *Let  $R$  be a principal ideal domain, and let  $M$  be a finitely generated  $R$ -module. Then*

$$M \cong R^s \oplus R/Ra_1 \oplus \dots \oplus R/Ra_r,$$

where  $s, r \in \mathbb{N} \cup \{0\}$ , the  $a_i$  are non-zero non-units in  $R$ , and  $a_i \mid a_{i+1}$  for  $i = 1, \dots, r-1$ .

*Proof.* Since  $M$  is finitely generated, by Lemma 5.24 there is a surjective  $R$ -homomorphism  $\phi_0 : R^n \rightarrow M$ , so  $M \cong R^n/K$  where  $K = \ker \phi_0$ . Since  $K$  is a submodule of the free module  $R^n$ , by Lemma 5.25  $K$  is free of rank  $m \leq n$ . Let  $\phi : R^m \xrightarrow{\sim} K$  be an isomorphism and  $\{e_1, \dots, e_m\}$  the standard basis of  $R^m$ . Writing  $\phi(e_j) = (a_{1j}, \dots, a_{nj})^\top \in R^n$ , the map  $\phi$  is left-multiplication by the matrix  $A = (a_{ij}) \in M_{n \times m}(R)$ , so  $K = AR^m$ .

By Theorem 8.8 (Smith Normal Form), there exist invertible  $P \in M_n(R)$  and  $Q \in M_m(R)$  such that  $PAQ = \text{diag}(a_1, \dots, a_k, 0, \dots, 0)$  with  $a_1 \mid \dots \mid a_k$ . Since  $Q$  is invertible,  $\{Qv \mid v \in R^m\} = R^m$ , so  $AR^m = AQR^m$ . Left-multiplication by the invertible  $P$  is an automorphism of  $R^n$  sending  $K = AQR^m$  to  $P(AQR^m) = PAQR^m$ , so  $R^n/K \cong R^n/PAQR^m$ . Therefore

$$M \cong \frac{R^n}{PAQR^m} \cong \frac{R}{Ra_1} \oplus \dots \oplus \frac{R}{Ra_k} \oplus R^{n-k},$$

where the last isomorphism is by Theorem 4.33, using that the image of  $R^m$  under  $\text{diag}(a_1, \dots, a_k, 0, \dots, 0)$  is  $Ra_1 \oplus \dots \oplus Ra_k \oplus 0 \oplus \dots \oplus 0 \subseteq R^n$ . Deleting the zero summands



(those where  $a_i$  is a unit, giving  $R/Ra_i = 0$ ), renaming the remaining non-unit diagonal entries  $a_1, \dots, a_r$ , and setting  $s = n - k$  gives

$$M \cong \frac{R}{Ra_1} \oplus \cdots \oplus \frac{R}{Ra_r} \oplus R^s. \quad \square$$

**Definition 26** (9.2 — Torsion element). *Let  $R$  be a principal ideal domain and  $M$  an  $R$ -module. An element  $x \in M$  is called a torsion element if there exists a non-zero  $r \in R$  such that  $rx = 0$ .*

**Definition 27** (9.4 — Torsion-free element). *Let  $R$  be a principal ideal domain and  $M$  an  $R$ -module. A non-zero element  $x \in M$  is called a torsion-free element if  $rx = 0$  for  $r \in R$  implies  $r = 0$ .*

**Theorem 52** (9.6). *Let  $R$  be a principal ideal domain and  $M$  an  $R$ -module. Then*

$$\text{Tor } M := \{x \in M \mid x \text{ is a torsion element}\}$$

*is a submodule of  $M$ .*

*Proof.* We verify the three conditions of Definition 4.8.

*Non-empty.* For any non-zero  $r \in R$ ,  $r \cdot 0 = 0$ , so  $0 \in \text{Tor } M$ .

*Closed under subtraction.* Let  $x, y \in \text{Tor } M$ , say  $rx = 0$  and  $sy = 0$  for non-zero  $r, s \in R$ . Then

$$rs(x - y) = s(rx) - r(sy) = 0.$$

Since  $R$  is a PID it is an integral domain, so  $rs \neq 0$ , hence  $x - y \in \text{Tor } M$ .

*Closed under scalar multiplication.* Let  $x \in \text{Tor } M$  with  $rx = 0$  for some non-zero  $r \in R$ , and let  $a \in R$ . Then  $r(ax) = a(rx) = 0$ , so  $ax \in \text{Tor } M$ .  $\square$

**Theorem 53** (9.7). *Let  $M$  be a finitely generated module over a principal ideal domain  $R$ . Then*

$$M = F \oplus \text{Tor } M,$$

*where  $F \cong R^s$  for some non-negative integer  $s$ , and  $\text{Tor } M \cong R/Ra_1 \oplus \cdots \oplus R/Ra_r$  with the  $a_i$  non-zero non-units satisfying  $a_1 \mid \cdots \mid a_r$ .*

*Proof.* By Theorem 9.1,  $M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_r$ . Set  $F \cong R^s$  and  $T \cong R/Ra_1 \oplus \cdots \oplus R/Ra_r$ , so  $M = F \oplus T$ . We show  $T = \text{Tor } M$ .

$T \subseteq \text{Tor } M$ . Since  $a_1 \mid a_2 \mid \cdots \mid a_r$ , we have  $a_r \in Ra_i$  for each  $i$ , so  $a_r$  annihilates every summand  $R/Ra_i$ . Hence  $a_r T = 0$ , meaning every element of  $T$  is torsion, so  $T \subseteq \text{Tor } M$ .

$\text{Tor } M \subseteq T$ . Let  $x \in \text{Tor } M$  and write  $x = x_1 + x_2$  with  $x_1 \in F$  and  $x_2 \in T$ . Since  $T \subseteq \text{Tor } M$ , the element  $x_2$  is torsion, and therefore so is  $x_1 = x - x_2$ . Say  $rx_1 = 0$  for some non-zero  $r \in R$ . Under the isomorphism  $F \cong R^s$ ,  $x_1$  corresponds to some  $(y_1, \dots, y_s) \in R^s$ , and  $r(y_1, \dots, y_s) = 0$  gives  $ry_i = 0$  for each  $i$ . Since  $R$  is an integral domain and  $r \neq 0$ , each  $y_i = 0$ , so  $x_1 = 0$ . Thus  $x = x_2 \in T$ .  $\square$

**Theorem 54** (9.8). *Let  $R$  be a principal ideal domain. Then any non-zero ideal  $P \neq R$  is prime if and only if it is maximal.*

*Proof.* ( $\Leftarrow$ ) Every maximal ideal is prime by Theorem 6.21.

( $\Rightarrow$ ) Let  $P = (a)$  be a non-zero prime ideal. Suppose for a contradiction that  $P$  is not maximal, so there exists an ideal  $M$  with  $P \subsetneq M \subsetneq R$ . Since  $R$  is a PID,  $M = (b)$

for some  $b \in R$ . Then  $(a) \subseteq (b)$  implies  $a = bx$  for some  $x \in R$ . Since  $b \notin (a)$  and  $(a)$  is prime, the relation  $a = bx \in (a)$  forces  $x \in (a)$ , say  $x = ay$  for some  $y \in R$ . Then  $a = bx = bay$ , and since  $R$  is an integral domain and  $a \neq 0$ , we may cancel to get  $1 = by$ . But then  $b$  is a unit, so  $M = (b) = R$ , a contradiction.  $\square$

**Corollary 7.** *Let  $R$  be a principal ideal domain and  $P$  a non-zero prime ideal. Then  $R/P$  is a field.*

*Proof.* Since  $P$  is a non-zero prime ideal in the PID  $R$ , Theorem 9.8 gives that  $P$  is maximal. It remains to show  $R/P$  is a field, i.e. that every non-zero element  $a + P \in R/P$  is invertible. Since  $a \notin P$ , the ideal  $Ra + P$  strictly contains  $P$ . As  $P$  is maximal,  $Ra + P = R$ , so there exist  $r \in R$  and  $p \in P$  with  $ra + p = 1$ . Then  $(r + P)(a + P) = ra + P = 1 + P$ , so  $a + P$  is invertible.  $\square$

**Theorem 55** (9.9 — Uniqueness of Decomposition). *Let  $R$  be a principal ideal domain and  $M$  a finitely generated  $R$ -module. Suppose*

$$M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_u \quad \text{and} \quad M \cong R^t \oplus R/Rb_1 \oplus \cdots \oplus R/Rb_v,$$

where the  $a_i$  and  $b_j$  are non-zero non-units with  $a_1 \mid \cdots \mid a_u$  and  $b_1 \mid \cdots \mid b_v$ . Then  $s = t$ ,  $u = v$ , and  $Ra_i = Rb_i$  for all  $i$ .

*Proof.* *Step 1:*  $s = t$ . By Theorem 9.7, both decompositions yield  $M = F \oplus \text{Tor } M$  with  $F \cong R^s$  and  $F' \cong R^t$  respectively. Then

$$R^s \cong F \cong M/\text{Tor } M \cong F' \cong R^t,$$

so  $s = t$  by Theorem 5.21. Let  $T = \text{Tor } M$ ; it remains to show  $u = v$  and  $Ra_i = Rb_i$ .

*Step 2:*  $u = v$ . For an  $R$ -module  $X$  and a prime  $p \in R$ , define  $X_p := \{x \in X \mid px = 0\}$ . By the corollary to Theorem 9.8,  $R/Rp$  is a field, so  $T_p$  is a vector space over  $R/Rp$ . From the first decomposition of  $T$ ,

$$\left( \frac{R}{Ra_i} \right)_p = \begin{cases} R/Rp & \text{if } p \mid a_i, \\ 0 & \text{if } p \nmid a_i, \end{cases}$$

so  $\dim_{R/Rp}(T_p)$  equals the number of  $a_i$  divisible by  $p$ . Now choose a prime  $p$  dividing  $a_1$ ; since  $a_1 \mid a_i$  for all  $i$ ,  $p$  divides every  $a_i$ , giving  $\dim_{R/Rp}(T_p) = u$ . Applying the same reasoning to the second decomposition, the dimension also equals the number of  $b_j$  divisible by  $p$ , which is at most  $v$ , so  $u \leq v$ . By symmetry (choosing a prime dividing  $b_1$ ),  $v \leq u$ , hence  $u = v$ .

*Step 3:*  $Ra_u = Rb_u$ . Since  $a_i \mid a_u$  for all  $i$ , the element  $a_u$  annihilates every summand in the first decomposition, so  $a_u T = 0$ . Applied to the second decomposition,  $a_u \cdot (R/Rb_u) = 0$ , which means  $a_u \in Rb_u$ , so  $Ra_u \subseteq Rb_u$ . By symmetry  $Rb_u \subseteq Ra_u$ , hence  $Ra_u = Rb_u$ .

*The remainder of the proof is non-examinable.* It remains to show  $Ra_i = Rb_i$  for  $i < u$  by downward induction on  $i$ . The idea is: assuming  $Ra_i = Rb_i$  for  $k \leq i \leq u$ , one multiplies both decompositions of  $T$  by a carefully chosen element  $x \in R$  built from a prime  $p$  and the exact  $p$ -adic valuation of  $a_u$ . This zeros out the first  $k - 1$  summands in one decomposition but not the other, producing two decompositions of  $xT$  with a different number of non-zero summands — contradicting the fact (established by the dimension argument in Step 2) that any two such decompositions must have equally many non-zero summands.  $\square$

**Theorem 56** (9.10 — Fundamental Theorem of Finitely Generated Abelian Groups). *Let  $A$  be a finitely generated abelian group. Then*

$$A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z},$$

where  $s \geq 0$ , the  $a_i$  are non-zero non-units in  $\mathbb{Z}$ , and  $a_1 \mid a_2 \mid \cdots \mid a_r$ . This decomposition is unique.

*Proof.* Every abelian group is a  $\mathbb{Z}$ -module (with  $n \cdot a$  meaning  $a + \cdots + a$  or its inverse), and  $\mathbb{Z}$  is a PID. So Theorem 9.1 applies directly with  $R = \mathbb{Z}$ , giving a decomposition  $A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z}$  where each  $\mathbb{Z}/a_i\mathbb{Z} = R/Ra_i$  is a cyclic torsion summand. Uniqueness of  $s$  and of the ideals  $\mathbb{Z}a_i$  (hence the  $a_i$  up to sign) then follows from Theorem 9.9.

If  $A$  is finite, then  $s = 0$ : the summand  $\mathbb{Z}^s$  contains elements of infinite order whenever  $s \geq 1$ , but a finite group has no such elements.  $\square$

## Lecture 13 — Rational Canonical Form

**Definition 28** (The  $F[x]$ -module structure on  $V$ ). *Let  $F$  be a field,  $V$  a finite-dimensional vector space over  $F$ , and  $T \in \text{Hom}_F(V, V)$  a linear map with matrix  $A$  relative to some basis. We make  $V$  into an  $F[x]$ -module by defining, for any polynomial  $f(x) = a_0 + a_1x + \cdots + a_mx^m \in F[x]$  and any  $v \in V$ ,*

$$f(x) \cdot v := a_0v + a_1(Tv) + \cdots + a_m(T^mv).$$

*This extends the  $F$ -action on  $V$  so that  $x \cdot v = Tv$ . Since any basis  $\{v_1, \dots, v_n\}$  of  $V$  over  $F$  also generates  $V$  as an  $F[x]$ -module,  $V$  is a finitely generated  $F[x]$ -module. Since  $F[x]$  is a PID (as  $F$  is a field), the structure theorem of Lecture 12 applies.*

**Theorem 57** (10.1). *Let  $T \in \text{Hom}_F(V, V)$ , and let  $f_1(x), \dots, f_n(x)$  be the invariant factors of  $A - xI$ , where  $A$  is a matrix of  $T$ . Then*

$$V \cong \frac{F[x]}{(f_1(x))} \oplus \cdots \oplus \frac{F[x]}{(f_n(x))}.$$

(*Proof non-examinable.*) The key steps are: define the surjective  $F[x]$ -homomorphism  $\phi : F[x]^n \rightarrow V$  by  $e_i \mapsto v_i$ , then show  $\ker \phi = (A - xI)F[x]^n$ , giving  $V \cong F[x]^n / (A - xI)F[x]^n$ . Since invertible changes of basis preserve this quotient up to isomorphism, one may apply Smith Normal Form (Theorem 8.8) to put  $A - xI$  in diagonal form  $\text{diag}(f_1(x), \dots, f_n(x))$ , after which Theorem 4.33 gives the direct sum decomposition.

**Remark 2** (10.2). *The polynomials  $f_1(x), \dots, f_n(x)$  are called the invariant factors of  $A$  (or of  $T$ ). After discarding any unit factors, the remaining non-unit invariant factors are called the invariant factors of the  $F[x]$ -module  $V$ .*

**Definition 29** (10.3 —  $T$ -cyclic space). *Let  $T \in \text{Hom}_F(V, V)$ . If  $V$ , regarded as an  $F[x]$ -module via  $T$ , is cyclic, then  $V$  is called a  $T$ -cyclic space and  $T$  is called a cyclic endomorphism of  $V$ .*

**Theorem 58** (10.4). *Let  $W$  be a submodule of the  $F[x]$ -module  $V$ , and let  $T \in \text{Hom}_F(V, V)$  satisfy  $TW \subseteq W$ . Then  $W$  is a  $T$ -cyclic subspace if and only if there exists  $w \in W$  such that  $\{w, Tw, \dots, T^{k-1}w\}$  is a basis of  $W$  for some  $k \in \mathbb{N}$ .*

*Proof.* ( $\Rightarrow$ ) Suppose  $W$  is  $T$ -cyclic, generated by  $w$ , so  $W = F[x] \cdot w$  in the sense of Definition 4.13. Since  $\dim V = n$  is finite, the vectors  $w, Tw, T^2w, \dots, T^nw$  must be linearly dependent over  $F$ . Let  $T^rw$  be the first of these linearly dependent on the preceding ones, so  $\{w, Tw, \dots, T^{r-1}w\}$  are linearly independent and

$$T^rw = a_0w + a_1(Tw) + \dots + a_{r-1}(T^{r-1}w), \quad a_i \in F.$$

We claim every  $T^iw$  is a linear combination of  $\{w, Tw, \dots, T^{r-1}w\}$ . This is clear for  $i \leq r-1$  and holds for  $i = r$  by the equation above. For  $i > r$ , apply  $T^{i-r}$  to the equation to get

$$T^iw = a_0(T^{i-r}w) + \dots + a_{r-1}(T^{i-1}w),$$

and each term on the right is expressible in terms of  $\{w, \dots, T^{r-1}w\}$  by the induction hypothesis. Since  $W = F[x] \cdot w$ , every element of  $W$  is of the form

$$f(x) \cdot w = (b_0 + b_1x + \dots + b_mx^m) \cdot w = b_0w + b_1(Tw) + \dots + b_m(T^mw),$$

using the  $F[x]$ -module action from the definition. Since every  $T^iw$  lies in  $\text{span}\{w, \dots, T^{r-1}w\}$ , this set spans  $W$ . Combined with linear independence,  $\{w, Tw, \dots, T^{r-1}w\}$  is a basis of  $W$  with  $k = r$ .

( $\Leftarrow$ ) Suppose  $\{w, Tw, \dots, T^{k-1}w\}$  is a basis of  $W$ . Any element  $c_0w + c_1(Tw) + \dots + c_{k-1}(T^{k-1}w)$  of  $W$  equals  $(c_0 + c_1x + \dots + c_{k-1}x^{k-1}) \cdot w \in F[x] \cdot w$  by the module action, so  $W \subseteq F[x] \cdot w$ . Since  $w \in W$  and  $W$  is closed under the  $F[x]$ -action,  $F[x] \cdot w \subseteq W$ . Hence  $W = F[x] \cdot w = (w)$ , so  $W$  is  $T$ -cyclic by Definition 10.3.  $\square$

**Definition 30** (10.5 — Companion matrix). *Let  $f(x) = a_0 + a_1x + \dots + a_{k-1}x^{k-1} + x^k$  be a monic polynomial over  $F$ . The companion matrix of  $f(x)$  is the  $k \times k$  matrix*

$$C(f) = \begin{pmatrix} 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & \dots & 0 & -a_1 \\ 0 & 1 & \dots & 0 & -a_2 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -a_{k-1} \end{pmatrix}.$$

**Definition 31** (10.7 — Minimal polynomial of  $T$  on  $W$ ). *Let  $W$  be a  $T$ -cyclic subspace generated by  $w$ . Since  $F[x]$  is a PID, the annihilator  $\text{Ann}(w) = \{f(x) \in F[x] \mid f(x) \cdot w = 0\}$  is a principal ideal  $(f(x))$  for a unique monic generator  $f(x)$ . This  $f(x)$  is called the minimal polynomial of  $T$  on  $W$ .*

**Theorem 59** (10.6). *Let  $T \in \text{Hom}_F(V, V)$  and let  $W$  be a  $T$ -cyclic subspace of  $V$  with basis  $\{w, Tw, \dots, T^{k-1}w\}$ . Then the matrix of  $T$  on  $W$  is the companion matrix of the minimal polynomial  $f(x)$  of  $T$  on  $W$ .*

*Proof.* Since  $\{w, Tw, \dots, T^{k-1}w\}$  is a basis of  $W$ , the vector  $T^kw$  is a linear combination of these basis elements, say

$$T^kw = a_0w + a_1(Tw) + \dots + a_{k-1}(T^{k-1}w), \quad a_i \in F.$$

In terms of the  $F[x]$ -module action ( $x \cdot v = Tv$ ), this says  $(a_0 + a_1x + \dots + a_{k-1}x^{k-1} - x^k) \cdot w = 0$ , so the polynomial  $g(x) := a_0 + a_1x + \dots + a_{k-1}x^{k-1} - x^k$  lies in  $\text{Ann}(w) = (f(x))$ .

Since  $F[x]$  is a PID,  $\text{Ann}(w)$  is a principal ideal  $(f(x))$  for some monic  $f(x)$  of degree  $s$ . The inclusion  $g(x) \in (f(x))$  gives  $s \leq k$ . On the other hand,  $f(x) \cdot w = 0$  means  $T^s w$

is a linear combination of  $\{w, Tw, \dots, T^{s-1}w\}$ . Since  $\{w, Tw, \dots, T^{k-1}w\}$  are linearly independent, this forces  $s \geq k$ . Hence  $s = k$ .

Since  $g(x) \in (f(x))$ , write  $g(x) = c \cdot f(x)$  for some  $c \in F$ . Both  $g$  and  $f$  are monic of degree  $k$  with leading coefficient  $-1$  and  $1$  respectively, so  $c = -1$  and  $f(x) = -g(x) = x^k - a_{k-1}x^{k-1} - \dots - a_1x - a_0$ .

The matrix of  $T$  on  $W$  relative to  $\{w, Tw, \dots, T^{k-1}w\}$  has columns given by the images of each basis vector:  $T(T^i w) = T^{i+1}w$  for  $i < k-1$ , contributing 1s on the subdiagonal, and the last column records the coefficients  $a_0, \dots, a_{k-1}$  of  $T^k w$ . By Definition 10.5, this is exactly the companion matrix of  $f(x)$ .  $\square$

## Lecture 14

**Theorem 60** (10.8 — Rational Canonical Form). *Let  $V$  be a finite-dimensional vector space over a field  $F$  and let  $T \in \text{Hom}_F(V, V)$ . Then there exists a basis of  $V$ , as an  $F[x]$ -module, with respect to which the matrix of  $T$  is*

$$\begin{pmatrix} B_1 & & \\ & \ddots & \\ & & B_r \end{pmatrix},$$

where  $B_i$  is the companion matrix of a monic polynomial  $f_i(x)$ , for  $i = 1, \dots, r$ , such that  $f_1(x) \mid f_2(x) \mid \dots \mid f_r(x)$ . This matrix is called the rational canonical form of  $T$ , and it is unique.

*Proof. Step 1:  $V$  is a torsion  $F[x]$ -module.* Since  $\dim_F V = n$ , we have  $\dim_F \text{Hom}_F(V, V) = n^2$ . So the  $n^2 + 1$  elements  $I, T, T^2, \dots, T^{n^2}$  must be linearly dependent over  $F$ . Let  $T^k$  be the first element linearly dependent on the preceding ones, so

$$T^k = a_0 I + a_1 T + \dots + a_{k-1} T^{k-1}, \quad a_i \in F.$$

Applying both sides to any  $v \in V$  and using the  $F[x]$ -module action  $x \cdot v = Tv$  gives

$$(a_0 + a_1 x + \dots + a_{k-1} x^{k-1} - x^k) \cdot v = 0.$$

Since this polynomial is nonzero, every  $v \in V$  is a torsion element (Definition 9.2), so  $V = \text{Tor } V$ .

*Step 2: Decompose  $V$  using Theorem 9.1.*  $V$  is a finitely generated  $F[x]$ -module: any  $F$ -basis of  $V$  also generates  $V$  as an  $F[x]$ -module, since  $F[x]$  is a PID. Since  $V = \text{Tor } V$ , the free part in Theorem 9.1 vanishes, giving

$$V = V_1 \oplus \dots \oplus V_r,$$

where each  $V_i \cong F[x]/(f_i(x))$  is a cyclic  $F[x]$ -module and  $f_1(x) \mid f_2(x) \mid \dots \mid f_r(x)$ .

*Step 3: Represent  $T$  on each  $V_i$ .* Each  $V_i$  is cyclic as an  $F[x]$ -module, hence a  $T$ -cyclic subspace (Definition 10.3). By Theorem 10.6, there exists a basis  $\mathcal{B}_i$  of  $V_i$  with respect to which  $T|_{V_i}$  has matrix  $B_i$ , the companion matrix of  $f_i(x)$ .

*Step 4: Assemble.* Since  $V = V_1 \oplus \dots \oplus V_r$ , the union  $\mathcal{B} = \mathcal{B}_1 \cup \dots \cup \mathcal{B}_r$  is a basis of  $V$ , and the matrix of  $T$  with respect to  $\mathcal{B}$  is  $\text{diag}(B_1, \dots, B_r)$ .

*Uniqueness.* The uniqueness of the decomposition subject to the divisibility chain, established in Theorem 9.9, implies the polynomials  $f_i(x)$  — and hence the companion matrices  $B_i$  — are uniquely determined by  $T$ .  $\square$

**Definition 32** (11.1 — Bi-additive map). *Let  $M$ ,  $N$ , and  $P$  be abelian groups. A map  $f : M \times N \rightarrow P$  is called bi-additive if*

$$f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n) \quad \text{and} \quad f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$$

*for all  $m, m_1, m_2 \in M$  and  $n, n_1, n_2 \in N$ .*

**Definition 33** (11.3 — Balanced map). *Let  $M$  be a right  $R$ -module and  $N$  a left  $R$ -module. A map  $f : M \times N \rightarrow P$  is called balanced if it is bi-additive and satisfies*

$$f(mr, n) = f(m, rn)$$

*for all  $m \in M$ ,  $n \in N$ , and  $r \in R$ .*

**Definition 34** (11.5 — Tensor product). *A tensor product of a right  $R$ -module  $M$  and a left  $R$ -module  $N$  is a pair  $(T, \phi)$ , where  $T$  is an abelian group and  $\phi : M \times N \rightarrow T$  is a balanced map, satisfying the following universal property: for every abelian group  $P$  and every balanced map  $f : M \times N \rightarrow P$ , there exists a unique group homomorphism  $\bar{f} : T \rightarrow P$  such that  $\bar{f} \circ \phi = f$ .*

**Remark 3** (11.6 — Uniqueness of the tensor product). *If  $(T', \phi')$  is another tensor product of  $M$  and  $N$ , then there exists an isomorphism  $\alpha : T \rightarrow T'$  such that  $\phi' = \alpha \circ \phi$ . Thus the tensor product is unique up to isomorphism. We write  $M \otimes_R N$  for the tensor product and  $m \otimes n$  for  $\phi(m, n)$ .*

**Theorem 61** (11.2 — Existence and construction of  $M \otimes_R N$ ). *Let  $M$  be a right  $R$ -module and  $N$  a left  $R$ -module. The tensor product  $M \otimes_R N$  exists. Concretely, it is realised as  $T(M, N) = A(M, N)/B(M, N)$ , where  $A(M, N)$  is the free abelian group with basis  $M \times N$ , and  $B(M, N)$  is the subgroup generated by all elements of the form*

$$\begin{aligned} (m_1 + m_2, n) - (m_1, n) - (m_2, n), \\ (m, n_1 + n_2) - (m, n_1) - (m, n_2), \\ (mr, n) - (m, rn), \end{aligned}$$

*for all  $m, m_1, m_2 \in M$ ,  $n, n_1, n_2 \in N$ , and  $r \in R$ .*

*(Proof non-examinable.)*

## Lecture 15

**Theorem 62** (11.7). *Let  $M$  and  $N$  be right and left  $R$ -modules, respectively. Then for all  $m, m_1, m_2 \in M$ ,  $n, n_1, n_2 \in N$ , and  $r \in R$ :*

- (i)  $(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$ ,
- (ii)  $m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$ ,
- (iii)  $mr \otimes n = m \otimes rn$ .

*(Proof non-examinable.)* These follow directly from the fact that  $B(M, N)$  maps to zero under  $\phi$ , so the defining relations of  $B(M, N)$  become identities in  $M \otimes_R N$ .

**Theorem 63** (11.8). *Let  $M$  and  $N$  be right and left  $R$ -modules, respectively.*

(a) *For  $\lambda \in \mathbb{Z}$ ,  $m \in M$ ,  $n \in N$ :  $(\lambda m) \otimes n = \lambda(m \otimes n) = m \otimes \lambda n$ . In particular,  $0 \otimes n = 0 = m \otimes 0$  and  $(-m) \otimes n = -(m \otimes n) = m \otimes (-n)$ .*

(b) *Every element of  $M \otimes_R N$  is a finite sum  $\sum_i m_i \otimes n_i$  for  $m_i \in M$ ,  $n_i \in N$ .*

(Proof non-examinable.) For (a), fix  $n$ ; the map  $m \mapsto m \otimes n$  is a group homomorphism, so it commutes with  $\lambda$ -multiples. For (b), every element of  $A(M, N)$  is a finite  $\mathbb{Z}$ -linear combination  $\sum \lambda_i(m_i, n_i)$ , and since  $\lambda_i(m_i \otimes n_i) = (\lambda_i m_i) \otimes n_i$  by (a), every element of  $M \otimes_R N$  is a finite sum of simple tensors.

**Remark 4** (Module structure on  $M \otimes_R N$  — 11.3). *If  $R$  is commutative, then  $M \otimes_R N$  can be given the structure of a right  $R$ -module by defining  $(m \otimes n)r = mr \otimes n$ . This is well defined because  $R$  being commutative ensures the map  $fr : (m, n) \mapsto mr \otimes n$  vanishes on  $B(M, N)$ .*

**Theorem 64** (11.9). *Let  $M$  be a right  $R$ -module. Then  $M \otimes_R R \cong M$  as additive abelian groups. If  $R$  is commutative, then  $M \otimes_R R \cong M$  as  $R$ -modules.*

(Proof non-examinable.) The map  $\bar{f} : M \otimes_R R \rightarrow M$  defined by  $\bar{f}(m \otimes r) = mr$  is a group isomorphism, with inverse  $g : m \mapsto m \otimes 1$ .

**Theorem 65** (11.10). *Let  $M$ ,  $N$ , and  $P$  be  $R$ -modules over a commutative ring  $R$ . Then*

(i)  *$M \otimes_R N \cong N \otimes_R M$  as  $R$ -modules,*

(ii)  *$(M \otimes_R N) \otimes_R P \cong M \otimes_R (N \otimes_R P)$  as  $R$ -modules.*

(Proof non-examinable.) For (i), the map  $m \otimes n \mapsto n \otimes m$  induces an isomorphism  $M \otimes_R N \rightarrow N \otimes_R M$ ; the inverse is constructed symmetrically. Part (ii) is analogous.

**Definition 35** (Tensor product of homomorphisms — §11.4). *Let  $f : M \rightarrow M'$  and  $g : N \rightarrow N'$  be  $R$ -homomorphisms. The tensor product  $f \otimes g : M \otimes_R N \rightarrow M' \otimes_R N'$  is the homomorphism defined by*

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n).$$

*When  $R$  is commutative,  $f \otimes g$  is an  $R$ -homomorphism.*

**Theorem 66** (11.12). *Let  $f : M \rightarrow M'$ ,  $f' : M' \rightarrow M''$ ,  $g : N \rightarrow N'$ ,  $g' : N' \rightarrow N''$  be  $R$ -homomorphisms. Then*

$$(f' \otimes g')(f \otimes g) = f'f \otimes g'g.$$

*In particular, if  $f$  and  $g$  are isomorphisms then  $f \otimes g$  is also an isomorphism.*

(Proof non-examinable.) Follows directly from the definition:  $(f' \otimes g')(f \otimes g)(m \otimes n) = f'f(m) \otimes g'g(n)$ .

**Theorem 67** (11.13). *Let  $(M_i)_{i \in I}$  and  $(N_j)_{j \in J}$  be families of right and left  $R$ -modules, respectively. Then*

$$\left( \bigoplus_{i \in I} M_i \right) \otimes_R \left( \bigoplus_{j \in J} N_j \right) \cong \bigoplus_{(i,j) \in I \times J} (M_i \otimes_R N_j),$$

*via the isomorphism  $(\sum_i x_i) \otimes (\sum_j y_j) \mapsto \sum_{i,j} x_i \otimes y_j$ .*

(*Proof non-examinable.*) The map  $f$  sending  $(\sum_i x_i, \sum_j y_j) \mapsto \sum_{i,j} x_i \otimes y_j$  is balanced, inducing a homomorphism  $\bar{f}$  in one direction. The inclusions  $\lambda_i : M_i \hookrightarrow \bigoplus M_k$  and  $\mu_j : N_j \hookrightarrow \bigoplus N_\ell$  give homomorphisms  $\lambda_i \otimes \mu_j$ , which extend to an inverse  $\bar{g}$ ; then  $\bar{f}$  and  $\bar{g}$  are mutually inverse isomorphisms.

**Theorem 68** (11.14). *Let  $R$  be a commutative ring,  $f : M \rightarrow M'$  and  $g : N \rightarrow N'$  be  $R$ -homomorphisms. If  $f$  and  $g$  are surjective, then  $f \otimes g$  is surjective.*

(*Proof non-examinable.*) Every simple tensor  $m' \otimes n' \in M' \otimes_R N'$  can be written as  $f(m) \otimes g(n) = (f \otimes g)(m \otimes n)$  since  $f$  and  $g$  are surjective; hence every generator of  $M' \otimes_R N'$  is in the image of  $f \otimes g$ .